MSc Business Information Technology
Research Topics

Redefining the SLR: An AI-Driven Review of Process Discovery in Automated Warehousing

Stefan de Laat BSc

Supervisors:
Dr. Faiza Bukhsh - University of Twente
Dr. Marcos Machado - University of Twente

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Faculty of Electrical Engineering, Mathematics and Computer Science (EEMCS),
Faculty of Behavioural, Management and Social sciences (BMS),
University of Twente

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Chapter 1

Introduction

The increasing automation of warehouses and intralogistics systems has fundamentally changed how material flows are executed, monitored, and managed. Modern automated logistics environments, such as high-throughput warehouses, sorting systems, automated storage and retrieval systems, and cross-docking facilities, rely on a dense network of control software, sensors, scanners, and embedded information systems to coordinate the movement of goods in real time. As a result, these environments continuously generate large volumes of event data that capture the operational behavior of the system at a highly detailed level. This growing availability of data creates significant opportunities for organizations to better understand process execution, improve transparency, and support operational and strategic decision-making [1].

At the same time, intralogistics systems are becoming more complex. The interaction between conveyors, shuttles, lifts, workstations, buffers, and software controllers produces process behavior that is often difficult to comprehend using traditional analysis methods alone. Managers and engineers therefore need methods that go beyond static reporting and conventional performance indicators, enabling them to uncover how processes are actually executed in practice, where deviations arise, and which structural bottlenecks reduce performance. In this context, data-driven analytical approaches are increasingly important for transforming raw operational data into actionable insights.

However, despite the availability of this event data, a disconnect remains between the actual execution of these processes and the theoretical frameworks used to manage them. Traditionally, Business Process Management (BPM) has relied on model-based analysis, utilizing handcrafted models that often represent an idealized view of reality: the "to-be" rather than the "as-is." This reliance on idealization frequently leads to management decisions based on "fiction rather than facts" [2]. Process mining emerges as the critical missing link that bridges the gap between these two disciplines.

As defined by Van der Aalst [2], process mining is an emerging discipline providing comprehensive sets of tools to provide fact-based insights and to support process improvements. Process mining techniques enable organizations to reconstruct the actual behavior of their processes by systematically exploiting event data recorded in information systems. The significance of this field is underscored by the *Process Mining*

Manifesto, which advocates treating event data as "first-class citizens" and distinguishes three main types of analysis: (1) *Process Discovery*, which infers process models from raw event logs; (2) *Conformance Checking*, which verifies the alignment between reality and models; and (3) *Process Enhancement*, which extends or enriches process models [3].

Within the context of intralogistics, process discovery is particularly relevant. Whereas conformance checking and enhancement require either an existing reference model or a previously discovered baseline, process discovery provides the essential first step: revealing the underlying structure of the actual process from recorded data. Applied to system logs, event logs, and sensor data, process discovery can expose the real flow of materials and activities through an automated logistics system. In doing so, it can possibly identify recurrent behavioral patterns, deviations, waiting points, loops, and bottlenecks that may otherwise remain hidden in highly complex technical environments.

Although process mining has matured in domains such as healthcare, ICT, and manufacturing, its application in (intra)logistics remains relatively fragmented [4]. Existing studies are distributed across related contexts such as warehousing, baggage handling, material handling, and distribution systems, and they often differ substantially in terms of data structure, discovery technique, and system scale.

This paper therefore presents a Systematic Literature Review (SLR) on process mining in warehouses and other types of intralogistics, with a specific focus on process discovery. The aim is to synthesize the current body of knowledge regarding which discovery techniques have been applied, how event data should be structured and preprocessed, which challenges are repeatedly encountered, and what kinds of insights process discovery can provide for automated logistics environments.

1.1 Research Questions

To address this objective in a structured manner, the review is organized around four research questions. Together, these questions examine the methodological, technical, and practical dimensions of applying process discovery in intralogistics. They are intended to provide a comprehensive overview of the state of the art, while also identifying gaps and opportunities for future research.

L1. What process discovery techniques have been applied in warehouse, intralogistics, or other automated material handling environments?

L2. What data structures and preprocessing methods are recommended for event logs, system logs, or sensor data in these environments for effective process discovery?

L3. What are the common challenges encountered when applying process mining techniques to large-scale intralogistics systems?

L4. What types of insights can process discovery provide for improving transparency, system understanding, and decision-making in automated logistics operations?

Chapter 2

Theoretical background

A SLR is a rigorous, methodical approach for identifying, evaluating, and synthesizing existing research on a specific topic [5]. It involves defining explicit criteria for selecting studies, critically analyzing the findings, and aggregating evidence to provide a comprehensive overview of the state of research. This review adopts a newly proposed AI-Augmented version of the methodological framework proposed by Okoli [5] to ensure a faster, more transparent and replicable review process for the topic of process mining in warehousing environment.

2.1 Methodology Proposal

This research proposes a novel adaptation of Okoli's traditional eight-step guide, re-engineered to leverage the capabilities of Large Language Models (LLMs), called *The AI-Augmented Systematic Literature Review*. As illustrated in Figure 2.2 in comparison to 2.1, the primary structural deviation involves switching the original Step 5 (Data Extraction) and Step 6 (Quality Appraisal), highlighted by the green dotted area.

In Okoli's methodology, manual quality appraisal is a labor-intensive and time-consuming task. Consequently, it is typically performed after data extraction to ensure reviewers only appraise a finalized subset of data. However, given the speed and efficiency with which AI can assess the relevance and rigor of a study, this framework performs quality appraisal first. Using an iterative AI-driven protocol, the model can filter studies based on predefined quality criteria before the extraction phase. This optimization ensures that comprehensive data extraction is only performed on relevant papers.

Also, the methodology introduces a bifurcated approach to Step 5 (Data Extraction). Extraction is divided into *metadata* and *knowledge* streams.

- **Metadata:** such as authors origins and publish date of the paper, is retrieved using the same type of iterative AI prompts used in the appraisal phase.
- **Knowledge:** remains a human-centric task to ensure depth of understanding. Crucially, this manual reading is supported by the AI's generated reasoning and summary.

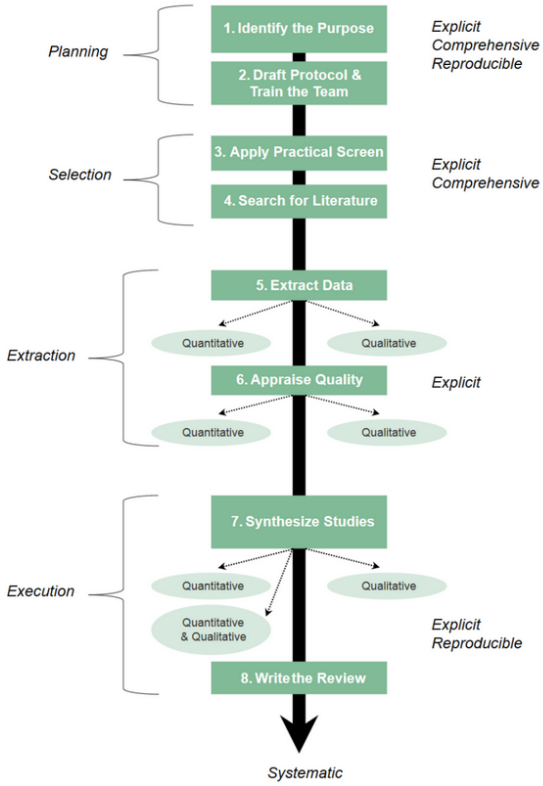

FIGURE 2.1: The Traditional SLR Process [5]

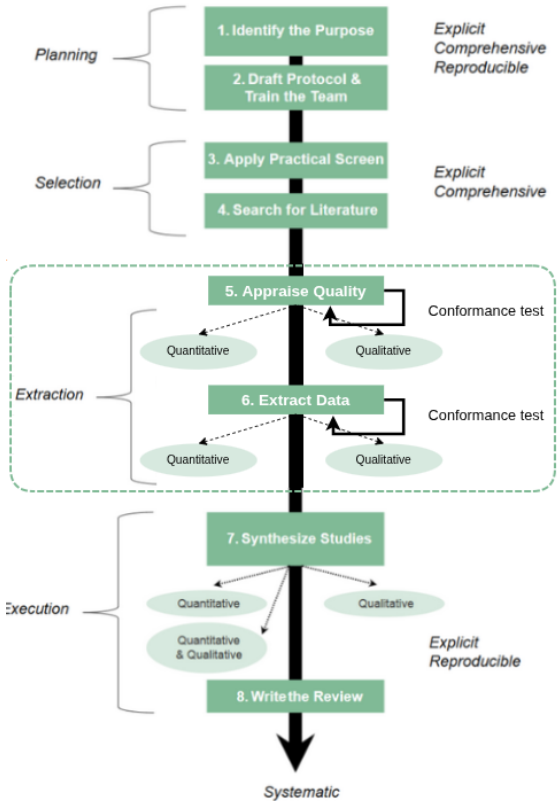

FIGURE 2.2: The AI-Augmented Systematic Literature Review process.

2.1.1 Consistency and Reproducibility

Unlike human reviewers, who are susceptible to fatigue and cognitive drift, leading to variations in judgment, an AI model can apply inclusion and exclusion criteria with deterministic consistency across the entire corpus. This addresses key challenges in manual screening:

- **Human Reviewers:** Multiple human reviewers are prone to assigning different relevance levels to the same paper, resulting in lower reproducibility of the screening process.
- **AI Model:** With the right settings, AI provides the exact same answer for the same input every time. This ensures high reproducibility and consistency, making the screening process more objective.

2.1.2 Speed and Scalability

The ability of AI systems to process and classify large volumes of textual data enables rapid screening of scientific literature at a scale that is impractical for manual reviewers. By expanding the pool of screened papers, the AI-augmented method improves the probability of identifying otherwise overlooked studies that may not be included in smaller manual reviews, which are limited in time.

2.1.3 Validity and Conformance

To ensure validity, this step employs a newly proposed conformance checking loop (see Figure 2.3), where the AI's scoring is validated against human judgment. If the accuracy of the predictions (for example the relevance label (step 5) and metadata (step 6)) is too low, the prompt logic is refined to better align with the researcher's vision and the batch is re-processed iteratively until the model's reasoning aligns with the inclusion criteria of the researcher, thus combining algorithmic speed with human oversight. Step 2 (draft protocol and train the team) and step 3 (apply practical screen) of Okoli's original methodology are still important, such that multiple researchers can contribute to validating the conformance checking process.

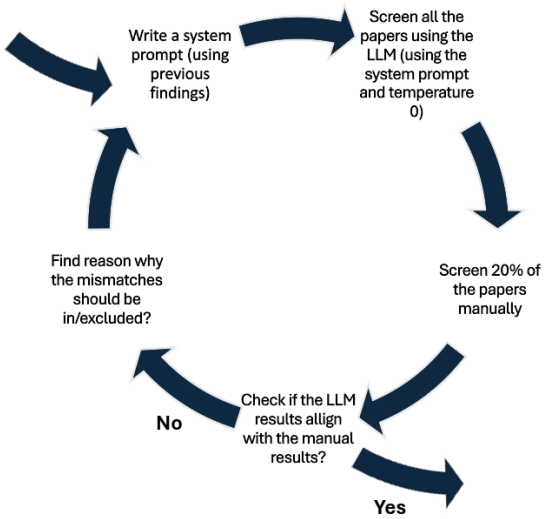

FIGURE 2.3: Conformance Checking Loop

2.2 The AI-Augmented Systematic Literature Review

This methodology section corresponds to steps 2 and 3 of Okoli's adapted framework, while step 1 (defining the research purpose and questions) has already been addressed in the introduction (see Section 1.1).

2.2.1 Search Query Development

The initial search query was developed using *Scopus* as the primary reference, with equivalent adaptations made for other academic databases. The goal of the query was to identify publications that discuss **process mining or process discovery** within the context of **logistics, warehousing, or material handling systems**, and that utilize **event or sensor data**.

The first version of the query was formulated as follows:

> ("process mining" OR "process discover*") AND ("warehous*" OR "intralogistic*" OR "airport cargo" OR "baggage handl*" OR "postal system" OR "sorting system" OR "material handl*" OR "cross-dock*" OR "ASRS" OR "AS/RS") AND ("event log*" OR "system log*" OR "sensor data" OR "tracking data")

This query aimed to ensure the retrieved studies were focused on process mining or discovery, applied in relevant industrial use cases, and utilizing similar data types. However, this initial formulation was too broad, yielding a large number of results, including many irrelevant papers where only one of the search terms appeared, often in the bibliography rather than the main content.

To improve precision, the search scope was refined to include only occurrences in the title, abstract, or keywords. The revised Scopus query is shown below:

> TITLE-ABS-KEY(("process mining" OR "process discover*") AND ("warehous*" OR "intralogistic*" OR "airport cargo" OR "baggage handl*" OR "postal system" OR "sorting system" OR "material handl*" OR "cross-dock*" OR "ASRS" OR "AS/RS") AND ("event log*" OR "system log*" OR "sensor data" OR "tracking data"))

This refinement significantly reduced irrelevant results and improved the focus of the literature retrieval process.

2.2.2 Database Selection

The literature search was conducted using three major academic databases: IEEE Xplore[1], Scopus[2], and Web of Science[3]. These well-known databases were selected due to their extensive coverage of peer-reviewed, high-quality research publications and their use in SLR guide within the field of Computer Science [6]. The combined use of these databases ensured comprehensive coverage of relevant studies while maintaining a high standard of academic reliability.

2.2.3 Inclusion and Exclusion Criteria

To ensure consistency and relevance, explicit inclusion and exclusion criteria will be applied during the selection process. These are based on the screening criteria guidelines provided by [5, chap. 4.1] and summarized in Table 2.1.

Table 2.1: Inclusion and Exclusion Criteria

Inclusion Criteria	**Exclusion Criteria**
Full-text screening by an LLM	Non-English publications
Peer-reviewed papers from reliable sources	Full text not available (including lack of access via the University of Twente subscription)
	Duplicate entries
	Complete conferences

2.2.4 Full-Text Screening

To full-text screen the remaining papers, OpenAI's GPT-4.1 mini[4] was used to determine whether a paper was relevant, together with a reason and a short summary of the paper. The temperature of the model was set to 0 (see Section 2.2.5), which in theory made the model deterministic. This had the advantage that the screening process was more reproducible than when performed by researchers, because the LLM produced the same answer each time, whereas different researchers could assign different relevance judgments to the same paper given the same research questions.

The next step was to test whether the model was reproducible by checking, for multiple papers, whether the model produced the same answer when the same prompt was used.

If the model proved to be reproducible, the final step was to check whether the model's answers aligned with the researcher's judgment. Therefore, a random selection

[1] https://ieeexplore.ieee.org
[2] https://www.scopus.com
[3] https://www.webofscience.com
[4] https://developers.openai.com/api/docs/models/gpt-4.1-mini

of 20% of the papers, maintaining the same relevant–irrelevant ratio, was selected by a Python program. These papers were manually checked for relevance and compared with the model's output. If the alignment was not sufficient (<85%), the prompt was adjusted to better reflect what was considered important and what the model had missed. This step was then repeated, making the process iterative

2.2.5 Use of temperature scaling

To ensure deterministic model behavior, a temperature setting of 10^{-12} was employed instead of 0. The temperature parameter is part of the softmax formulation used to convert model logits into a probability distribution over possible tokens. The generalized softmax function with temperature scaling is defined as [7]:

$$S(x_i) = \frac{\exp(x_i/\tau)}{\sum_{j=1}^{N} \exp(x_j/\tau)}, \tag{2.1}$$

where τ denotes the temperature parameter controlling the smoothness of the resulting probability distribution. As τ increases, the output distribution becomes more uniform, increasing randomness during token sampling. Conversely, as τ approaches zero, the distribution becomes increasingly peaked. In the limit $\tau \to 0$, the softmax function assigns a probability of 1 to the element with the highest value and a probability of 0 to all others [7]. In such a case, the most probable token would always be selected, resulting in deterministic output where the same sequence of tokens is generated for identical inputs.

However, the formulation of the softmax function shows that the temperature parameter appears in the denominator of the exponent. Setting $\tau = 0$ would therefore result in an undefined expression, meaning that a true zero temperature cannot be used in practice. Consequently, when a value of 0 is provided to the OpenAI API, it is likely interpreted as a placeholder and internally replaced by a non-zero value. Experimental observations indicate that this internally used value is still sufficiently large to introduce variability in the sampling process, as outputs produced with a temperature of 0 remained somewhat unpredictable. To better approximate deterministic behavior, a very small non-zero temperature of 10^{-12} was therefore used in this work, which resulted in significantly more consistent outputs. Additional empirical evidence supporting this observation is provided in Appendix A.

Chapter 3

SLR Study

3.1 Methodology results

This results section corresponds to steps 4, 5 & 6 of Okoli's adapted framework.

3.1.1 Paper Selection

The initial collection of papers was compiled from multiple databases. Entries were filtered as visualized in Figure 3.1. It is noteworthy that the World of Science database did not contribute any unique sources, as all papers identified in World of Science were already present in the results from the other two databases. Therefore, it was excluded from further consideration.

Using the defined system prompt, all papers were systematically reviewed by OpenAI's GPT-4.1 mini model with the temperature set to 10^{-12}. For each paper, the model assigned a binary relevance label, provided a brief justification for this decision, and generated a summary of the paper's content.

To assess conformance and reliability, a random sample of nine papers (representing 20% of the total dataset) was manually reviewed. The detailed results of this verification are presented in Appendix B. The model achieved an accuracy of 88.9%, a precision of 100%, and a recall of 83.3%. Importantly, no relevant papers were excluded, and only non-relevant papers were erroneously included. Thus, no essential information was omitted from the final selection.

A overview of the used Python code can be found on Github[1] and a comprehensive overview of all included and excluded papers, along with their with AI assessed relevance to this study, the reason why and a summery are provided in E.

[1] https://github.com/SirStepheno/ai-slr-automation-framework

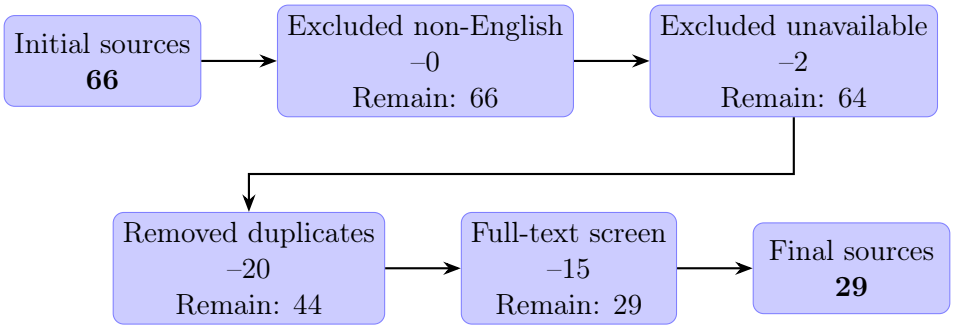

FIGURE 3.1: Filtering and selection process of papers

3.1.2 Statistics

This section provides an overview of key bibliographic characteristics of the collected literature. The extraction of these metadata—such as publication year and country of researcher affiliation, was performed using the same procedure applied during the relevance assessment described above. The only change was the use of an alternative system prompt tailored specifically to identify publication years and author affiliations. As with the relevance evaluation, a random sample of 20% of the papers was manually verified to assess accuracy. The validation results, presented in Appendix C, confirmed that the model performed reliably for these tasks as well. Based on this satisfactory performance, the extracted metadata were accepted as accurate and were used to generate the analyses in the following subsections. A complete overview of all extracted data can be found in Appendix D.

Publication Years

The temporal distribution of publications reveals how academic attention to the topic has evolved over time. As illustrated in Figure 3.2, research output spans the period from 2006 to 2025. Although publication frequency peaked in 2020, interest in the subject has remained steady, with new papers continuing to appear in recent years. This indicates that the field is still active and of current relevance.

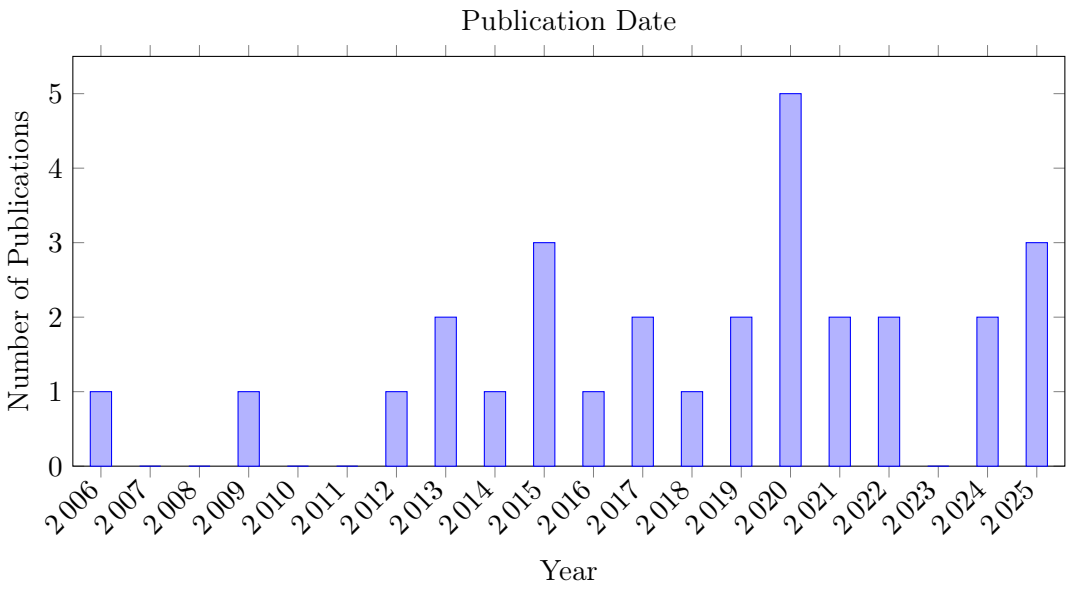

FIGURE 3.2: Publication years of included papers

Geographical Distribution of Research Activity

The geographical distribution of contributions was determined based on the institutional affiliations of the researchers involved in each publication. Because the number of authors varies across papers, a weighted scoring method was applied: each author contributes an equal fraction of one point to their country. For instance, a paper with 4 different authors assigns a score of 0.25 to each author's country of affiliation. When author has multiple affiliations, the first one was used for this statistic. This approach ensures a more balanced representation of research contributions.

The resulting global map (Figure 3.3) displays all countries represented in a world map, with non-contributing countries shown in gray. Additionally, Figure 3.4 presents the six most important countries with a cumulative score greater than 1, highlighting the countries with the greatest commitment to this area of research.

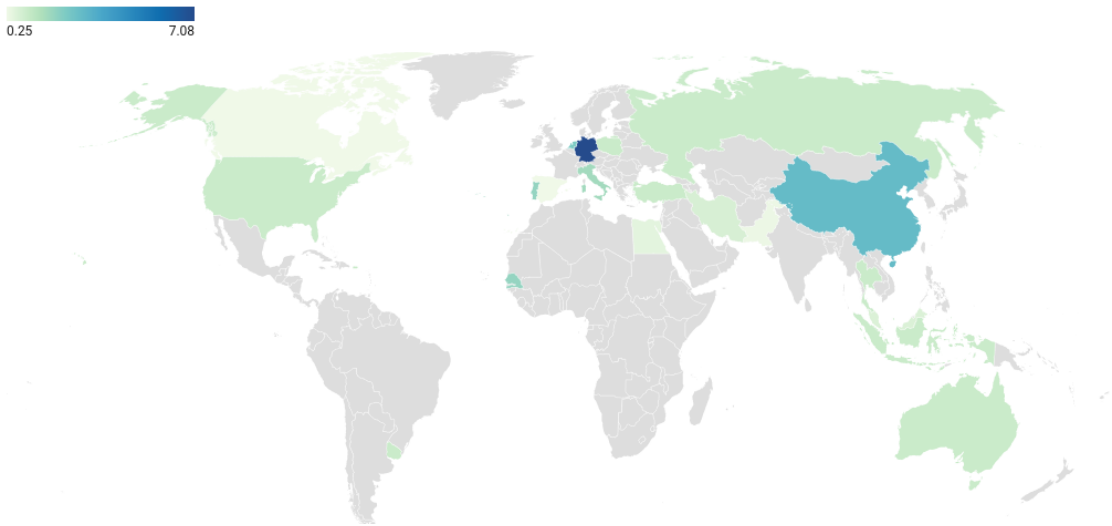

FIGURE 3.3: World Map showing geographical distribution of research activity

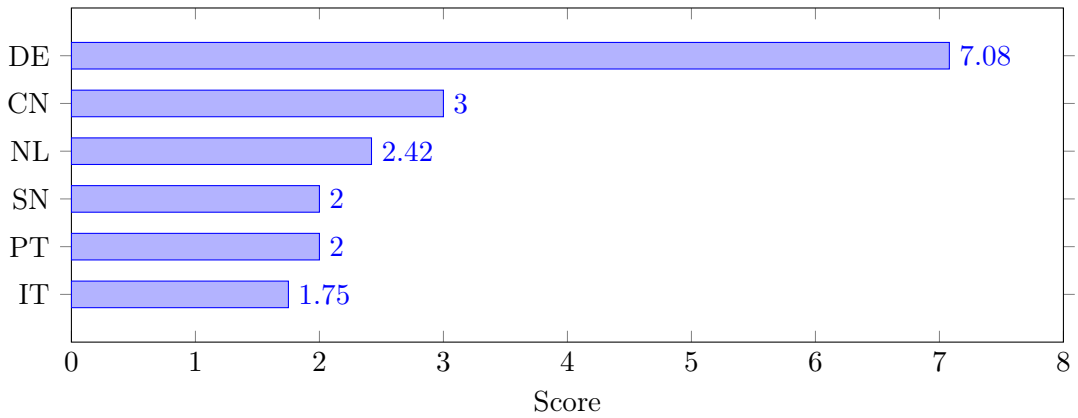

FIGURE 3.4: Countries with weighted research contribution score above 1

3.2 SLR results

This section presents the results of the Systematic Literature Review (SLR). The findings are structured around the research questions introduced earlier (see Section **??**) in this study and correspond to steps 7 and 8 of Okoli's adapted SLR framework, namely data extraction and synthesis. Each subsection addresses one research question, synthesizing and comparing insights from the selected literature to provide a structured overview of existing approaches, challenges, and contributions related to process discovery in automated warehousing and intralogistics environments.

L1. What process discovery techniques have been applied in warehouse, intralogistics, or other automated material handling environments?

Process discovery in automated material handling and warehousing environments has been addressed through a combination of classical algorithms, commercial software tools, and novel structural.

To manage the complexity and noise inherent in real-world logistical data, the Heuristic Miner is frequently utilized; it has been applied to model material movements in a shoe manufacturer's warehouse to identify deviations from standard procedures [8], and utilized alongside the Inductive Miner to optimize resource allocation in cross-docking systems [9].

Open-source frameworks and commercial software suites are the primary drivers of process mining analysis in this domain. Within the open-source ProM framework, Heuristic and Fuzzy Miners have been employed to visualize process sequences and identify stray signals [10], while the α-algorithm has been utilized to construct Petri nets for consistency checking in raw material warehouses [11].

Simultaneously, commercial tools play a significant role in industrial applications. Disco is widely used for rapid insight generation and conformance checking in diverse settings, ranging from scientific laboratory equipment facilities [12] to mattress inventory processes [13]. Furthermore, recent studies have leveraged Microsoft Power Automate in combination with Object-Centric Process Mining to capture the interactions of multiple objects in intralogistics [14].

Beyond standard control-flow discovery, specialized (visualization) techniques have been developed to address specific logistical characteristics: the "From-To Chart" method facilitates the identification of most common immediate successors [15], and context-aware approaches utilizing k-medoids clustering have been introduced to categorize heterogeneous process variants based on frequency and cycle time stability [16]. Furthermore, in high-performance environments like baggage handling systems, discovery efforts focus on "system-level" behaviors, utilizing the Performance Spectrum to map performance measures over time [17] and detecting cascades of correlated events to identify dynamic bottlenecks and high-load situations [18].

L2. How can event logs, system logs, or sensor data from such environments be structured and preprocessed for effective process discovery?

Effective process discovery relies on the structuring of raw data into exploitable formats, such XES and OCEL files. Fundamentally, event logs must contain a case ID, resource, activity name, and timestamp to be viable for analysis [19]. To manage the complexity of modern processes, literature suggests integrating these logs into broader meta-models. Kassem [20] proposes connecting event logs to a system meta-model broken down to the smallest measurable unit to maintain a clear overview. This aligns with the model-driven approaches advocated by Delgado [21] and Themistocleous [22]. The latter introduces the Business and Process Analytics Data Warehouse Meta-Model (B&PADWM), which utilizes Fact and Dimension tables to integrate business KPIs with process perspectives.

For advanced analytical capabilities, Volgelgesang et al [23] suggests structuring data into Multidimensional Event Logs (MEL) within relational databases to support complex filtering and aggregation, a concept later further operationalized through the PMCube Explorer interface. Additionally, the structure of fact tables can vary based on the goal; Yoganguina [24] highlights the distinction between transactional fact tables and cumulative fact tables, noting that the latter capture the state of a process at specific points in time, which is particularly effective for predictive monitoring using neural networks.

Data extraction and integration often require Extraction-Transformation-Loading (ETL) processes. When facing unstructured or heterogeneous business data, Pospiech [25] proposes a formalism using components and connectors to trace and reconstruct process instances across multiple data stores. To enhance the quality and interpretability of this preparation phase, Delgado [26] presents EVErPREP, a workflow model that leverages Event Knowledge Graphs (EKG) to facilitate the exploration and processing of event data. In scenarios where event logs are unavailable, Benvenuti [27] suggests an interactive approach that utilizes questionnaires to gather domain knowledge, which is subsequently used to generate simulated event logs for discovery.

Once structured, the data must undergo preprocessing to ensure accuracy and enable higher-level analysis. Wunderlich [14] emphasizes the importance of filtering, noting that traces remaining inside the system after the observed time should be ignored. Conversely, Janicki [28] addresses the issue of incomplete data by proposing methods to repair event logs with missing events.

Beyond cleaning, preprocessing involves abstraction and clustering to derive deeper insights. Toosinezhad [18] introduces a method of abstracting data from the case level to the segment and system levels, allowing for the spatial and temporal linking of dynamic bottlenecks. Finally, Becker [16] demonstrates that automated clustering, specifically using k-medoids, can be performed based on the context of object characteristics within the logs to better categorize process variations.

L3. Which challenges are commonly encountered when applying process mining techniques to large-scale intralogistics systems?

Applying process mining to large-scale intralogistics is often hindered by incomplete, scattered, heterogeneous data, which requires extensive preprocessing to establish a standardized format. Wunderlich et al [14] highlight that the high complexity of logistics processes and the large number of interacting elements present significant difficulties, particularly regarding data granularity where logistical units are frequently combined and separated. Consequently, identifying and relating all log data to the smallest business unit proves to be a difficult task. This challenge is driven by the fact that attributes and their values are often technically separated despite being logically connected, creating significant issues regarding data heterogeneity [20, 29]. Furthermore, system logs are frequently scattered across different locations and formats [30], sometimes involving unstructured data that requires complex key mapping across the system [25]. As noted by [27], translating these disparate sources into a uniform log is time-intensive and costly, often requiring specific subject matter knowledge to perform the necessary preprocessing.

Beyond initial data preparation, the volume and completeness of the event data present distinct operational challenges. The huge amount of data generated by these systems, combined with the joining of attributes and different sources, can lead to poor system performance and reach computational limits [20].

Another issue is incomplete logging, which arises from various causes, such as the strategic minimization of sensors within the environment to reduce costs. To mitigate the resulting inconsistencies and the complexity of reconstructing event orders, a specific approache utilize the many-to-many interactions between cases and resources with shared resources and queues to repair event logs via token trajectories and Linear Programming (LP) constraints [28].

L4. What types of insights can process discovery provide for improving transparency, system understanding, and decision-making in automated logistics operations?

Process discovery offers a diverse range of insights that enhance transparency, deepen system understanding, and support strategic decision-making in automated logistics. By analyzing event logs, organizations can uncover discrepancies between actual operations and standard procedures. For instance, [8] utilize dotted charts for FIFO analysis to visualize material movement traces, revealing deviations that directly impact holding times and overall efficiency. Similarly, [12] demonstrate how analyzing event frequency via histograms can effectively detect anomalies in warehouse operations, such as significant variations in product inspection times. To manage the complexity of heterogeneous logistic data, [16] propose clustering techniques like k-medoids to identify main process variants, thereby enhancing transparency regarding process variations. Furthermore, [23] introduce multidimensional process mining, which allows for multiple perspectives on process variants to identify bottlenecks and differences, as demonstrated in healthcare case studies but applicable to logistics. These discovery techniques are fundamental

for establishing a baseline understanding of system behavior, although [14] caution that identified causes of inefficiencies should initially be treated as hypotheses requiring further validation.

Beyond structural analysis, process discovery facilitates advanced decision-making through the integration of performance metrics and predictive capabilities. [22] propose a meta-model that integrates Key Performance Indicators (KPIs) and fact tables directly into the process model, enabling the calculation of KPIs on the go, which can then be visualized in a dashboard, besides generation a process model. To assess the qualitative aspect of processes, [29] incorporate satisfaction scores as a KPI for individual traces, linking process execution to patient outcomes. Predictive insights are also achievable; [17] present a methodology using the performance spectrum to extract features for machine learning, predicting future system performance like workload and delay risks based on the current system state. Complementing this, [24] show that predictive models for delivery times allow employees to anticipate delays, optimize daily planning, and align operations with financial objectives, ultimately reducing supply chain costs.

At last, even in scenarios with limited process knowledge, [27] suggest using structured questionnaires to validate discovered models, ensuring that the insights generated are both accurate and actionable for decision-makers.

Chapter 4

Discussion

4.1 Replication and Reproducibility of the AI-Augmented SLR

A central contribution of this study lies in its emphasis on reproducibility within the SLR process. Traditional SLRs are inherently subject to human interpretation, which introduces variability in study selection and classification. In contrast, the proposed AI-augmented methodology demonstrates a high degree of determinism and consistency. By configuring the LLM with an extremely low temperature (10^{-12}), the screening process becomes effectively deterministic, ensuring that identical inputs consistently yield identical outputs. This characteristic directly addresses one of the long-standing challenges in SLR methodology: the reproducibility of inclusion and exclusion decisions.

The conformance checking loop further strengthens this claim. The observed precision of 100% and accuracy of 88.9% indicate that the LLM reliably aligns with human judgment while eliminating intra-reviewer variability.

In addition, the AI-augmented approach significantly improves the efficiency of the screening phase. Compared to manual review, the automated process enables the rapid evaluation of a substantially larger pool of candidate papers. This allows researchers to formulate less restrictive initial search queries, thereby increasing recall and reducing the risk of overlooking relevant studies.

Overall, the results demonstrate that the proposed methodology performs strongly in terms of consistency, reliability, and efficiency. However, despite these promising outcomes, the integration of AI into SLR methodologies remains an emerging research area. Several methodological and technological aspects require further investigation to fully validate and generalize the approach. These directions are discussed in more detail in Section 4.3.

4.2 Scientific Novelty and Practical Implications

The contributions of this study can be understood by synthesizing the findings of the research questions (L1–L4). Regarding L1, the study consolidates the application of process discovery techniques in intralogistics, demonstrating that classical algorithms such as the Heuristic Miner and Inductive Miner remain highly relevant when adapted to noisy and complex industrial data. This contributes to the scientific understanding of how established techniques are applied in modern automated environments. With respect to L2, this research highlights the critical role of data structuring and preprocessing, identifying meta-models, multidimensional event logs, and knowledge graph-based approaches as key enablers for effective analysis, thereby emphasizing the importance of semantic enrichment in process mining. Addressing L3, the study systematically identifies core challenges in large-scale intralogistics systems, including data heterogeneity, incompleteness, and scalability limitations, contributing a structured overview of recurring issues in the domain. Finally, in relation to L4, the study demonstrates that process discovery provides value beyond descriptive insights, enabling predictive analytics, KPI-driven decision-making, and enhanced system transparency, reflecting the evolution of process mining toward more advanced analytical capabilities.

In addition to these domain-specific contributions, the AI-augmented SLR methodology itself represents a methodological innovation. The domain-specific findings translate into several practical implications for practitioners in intralogistics:

- **Standardized data structuring**: Organizations should invest significantly in pre-mining data engineering efforts. Event data should be stored and structured in standardized formats so it can be easily converted to formats such as XES or OCEL, to ensure compatibility with process mining techniques and avoid costly reprocessing steps later.

- **Transition to open-source scalability and openness**: Practitioners should move away from black-box or GUI-based tools such as Disco or preconfigured ProM setups. Instead, leveraging open-source frameworks (e.g., PM4Py) allows for customizable, scalable solutions tailored to specific industrial requirements.

- **From visualization to actionable insights and control**: Process mining should extend beyond visualizing process flows. By incorporating contextual information (e.g., KPIs, system states, or semantic layers), organizations can derive meaningful insights and enable informed decision-making and process improvements.

4.3 Future Research Directions

First, there is a growing need for research focused on PM4Py (Process Mining for Python). While traditional tools like Disco and ProM are widely cited in the existing literature, they can face limitations regarding flexibility and scalability with massive industrial datasets. PM4PY is an upcoming, open-source library that offers significant advantages in terms of customizability and integration with broader data science pipelines. Future studies should explore the specific application of PM4Py in high-volume logistics environments, utilizing its algorithmic efficiency to handle event logs that exceed the capacity of standard (commercial) software.

Second, the integration of knowledge graphs and process mining represents a promising semantic frontier. Combining process mining with knowledge graphs allows for semantic abstraction, enabling the synthesis of unified event types. This enrichment facilitates "Drill-Down" and "Roll-Up" operations, allowing researchers to switch between low-level machine signals and high-level functional blocks. Further research into this synergy could significantly enhance the interpretability of process models in highly automated systems.

Third, further validation of the AI-augmented SLR methodology is required. While this study demonstrated strong alignment between the LLM and a single human reviewer, future research should involve multiple domain experts to establish a more robust baseline. This would enable the analysis of inter-rater reliability among human reviewers compared to the consistency of the LLM, providing deeper insights into the extent to which AI can reduce subjectivity in literature reviews.

Finally, future work should investigate the impact of different LLM architectures on the reproducibility and quality of SLR outcomes. This study utilized an older OpenAI model due to its explicit temperature controllability, which enabled deterministic behavior. However, newer reasoning-oriented models may offer improved contextual understanding while lacking such control mechanisms. Comparative studies are therefore needed to evaluate whether newer models can achieve similar levels of consistency, or whether deterministic configurations remain essential for reproducible AI-assisted SLR methodologies.

Chapter 5

Conclusion

This systematic literature review shows that process mining has clear potential to improve transparency, system understanding, and decision-making in intralogistics. The reviewed studies demonstrate that existing process discovery techniques can reveal process variants, bottlenecks, deviations, and performance-related patterns in logistics environments. At the same time, the literature confirms that successful application depends heavily on substantial data preparation, including the structuring, integration, and preprocessing of heterogeneous event data.

However, the review also reveals an important gap in the current state of research. Existing empirical studies are predominantly conducted in relatively small-scale or semi-manual distribution settings, or rely on simulated environments rather than real-world automated systems. Although related domains such as baggage handling, parcel sorting, postal logistics, and retail distribution provide useful reference points, they do not fully reflect the scale, density, and automation level of large contemporary distribution centres. For instance, a major European airport like Munich only process up to 50,000 bags per day [31]. As a result, empirical evidence on process mining in highly automated, high-volume intralogistics systems remains limited.

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Appendix A

Temperature Settings Conformance Checking

Table A.1: Different outputs gpt-4.1-mini for paper 29 with temperature 0

Relevant	Reason	Summary
TRUE	The paper directly addresses the application of process discovery techniques (heuristic miner and inductive miner algorithms) in an automated material handling environment, specifically a cross-docking warehouse system. It details the structuring and preprocessing of event logs collected via RFID systems, which aligns with RQ L2. The study discusses challenges such as deviations and process variations encountered in real-life warehouse operations, relevant to RQ L3. Furthermore, it demonstrates how process discovery combined with discrete-event simulation provides insights into process bottlenecks, resource utilization, and layout optimization, supporting RQ L4. The paper also exemplifies the use of event logs and sensor data for process mining in a warehouse setting, contributing to RQ L1. The methodology, case study, and results sections provide concrete evidence of these contributions.	This paper presents a case study on applying process mining and simulation to a newly established cross-docking warehouse system. Using RFID technology, detailed event logs capturing product flows and activities were collected and preprocessed to handle complexities such as product assembly. The authors applied heuristic and inductive miner algorithms to discover the actual process models, revealing deviations and inefficiencies, particularly in resource usage and process flows. Discrete-event simulation was then used to evaluate different scenarios, including resource allocation and door usage optimization, leading to recommendations for improved warehouse layout and operations. The study highlights the integration of process mining with simulation as a powerful approach to enhance transparency, system understanding, and decision-making in automated intralogistics environments.

Continued on next page

Relevant	Reason	Summary
TRUE	The paper directly addresses all four research questions of the SLR. For L1, it applies heuristic miner and inductive miner process discovery techniques to a cross-docking warehouse environment (Sections 4 and 5). For L2, it details the structuring and preprocessing of event logs derived from RFID system data, including integration of assembled products into single cases and enrichment of event logs with timestamps and resource usage (Section 4, Phase 2 and 5.1). For L3, it discusses challenges such as deviations in process flows, incomplete process executions, and data quality issues, and how these affect process mining and simulation (Sections 5.2 and 6). For L4, it demonstrates insights gained from process discovery and simulation, such as identifying underutilized doors, bottlenecks at specific stations, deviations from planned processes, and recommendations for layout optimization and resource allocation to improve transparency and decision-making (Sections 5.2 and 6). The paper also integrates process mining with discrete-event simulation to evaluate improvement scenarios, which is highly relevant to understanding and optimizing automated material handling systems.	This paper presents a case study on applying process mining and simulation to a newly established cross-docking warehouse system of a chain store. Using RFID technology, detailed event logs capturing product flows and activities were collected and preprocessed to handle complexities such as product assembly. The authors applied heuristic and inductive miner algorithms to discover the actual process models, revealing deviations and bottlenecks in the warehouse operations. They then used discrete-event simulation to evaluate different scenarios, including resource allocation and door usage optimization. The study provides actionable insights for improving warehouse layout, resource utilization, and process adherence, demonstrating the value of combining process mining with simulation for operational transparency and decision support in automated intralogistics environments.

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Relevant	Reason	Summary
TRUE	The paper directly addresses RQ L1 by applying process discovery techniques (heuristic miner and inductive miner algorithms) to a cross-docking warehouse environment, which is a form of automated material handling and intralogistics. It also contributes to RQ L2 by detailing the structuring and preprocessing of event logs collected via an RFID system, including integration of assembly and bundling processes into usable event logs. The paper discusses challenges related to deviations and process variations in the warehouse, thus informing RQ L3 about common issues such as process deviations, incomplete process executions, and resource allocation inefficiencies. Finally, it provides insights relevant to RQ L4 by demonstrating how process discovery combined with discrete-event simulation can improve transparency, identify bottlenecks, optimize resource allocation (e.g., number and location of doors), and support decision-making for warehouse layout and operations. The case study is comprehensive, grounded in real data, and includes practical recommendations for improving cross-docking operations, making it highly relevant for the SLR focused on automated material handling and intralogistics.	This paper presents a case study on applying process mining and simulation to a newly established cross-docking warehouse belonging to a chain store. Using RFID technology, detailed event logs capturing product flows and activities were collected and preprocessed to integrate complex processes such as assembly and bundling. The authors applied heuristic and inductive miner algorithms to discover the actual process models, revealing deviations and inefficiencies compared to the organizational ideal. They then used discrete-event simulation to evaluate different scenarios, including resource allocation and door usage optimization. The study identified underutilized doors that could be removed and highlighted stations with bottlenecks requiring additional resources. The results led to an improved warehouse layout and operational recommendations. The paper contributes a data-driven methodology combining process discovery and simulation to enhance transparency, system understanding, and decision-making in automated intralogistics environments.

Table A.2: Different outputs gpt-4.1-mini for paper 29 with temperature 10^{-12}

Relevant	Reason	Summary
TRUE	The paper directly addresses the application of process discovery techniques (heuristic miner and inductive miner algorithms) in an automated material handling environment, specifically a cross-docking warehouse system. It details the structuring and preprocessing of event logs collected via RFID systems, which aligns with RQ L2. The study discusses challenges such as deviations and process variations encountered in real-world warehouse operations, relevant to RQ L3. Furthermore, it demonstrates how process mining combined with discrete-event simulation provides insights into process flows, bottlenecks, resource usage, and layout optimization, supporting RQ L4. The paper also exemplifies the use of event logs and sensor data for process discovery in a warehouse setting, contributing to RQ L1. The methodology, results, and discussion sections provide concrete evidence of these contributions, making it highly relevant for the SLR focused on process discovery and mining in automated intralogistics.	This paper presents a case study on applying process mining and simulation to a newly established cross-docking warehouse system of a chain store. Using RFID technology, detailed event logs capturing product flows and activities were collected and preprocessed to handle complexities such as product assembly and bundling. The authors applied heuristic and inductive miner algorithms to discover the actual process models, revealing deviations and inefficiencies, particularly in certain stations and resource usage. Discrete-event simulation was then used to evaluate different scenarios, including resource allocation and door usage optimization, leading to recommendations for improved warehouse layout and operations. The study highlights the integration of process mining with simulation as a data-driven approach to enhance transparency, system understanding, and decision-making in automated material handling environments.

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Relevant	Reason	Summary
TRUE	The paper directly addresses the application of process discovery techniques (heuristic miner and inductive miner algorithms) in an automated material handling environment, specifically a cross-docking warehouse system. It details the structuring and preprocessing of event logs collected via RFID systems, which aligns with RQ L2. The study discusses challenges such as deviations and process variations encountered in real-life warehouse operations, relevant to RQ L3. Furthermore, it demonstrates how process discovery combined with discrete-event simulation provides insights into process flows, bottlenecks, resource usage, and layout optimization, supporting RQ L4. The paper also exemplifies the use of event logs and sensor data for process mining in a warehouse setting, contributing to RQ L1. The methodology, results, and discussion sections provide concrete evidence of these contributions.	This paper presents a case study on applying process mining and simulation to a newly established cross-docking warehouse system of a chain store. Using RFID technology, detailed event logs capturing product flows and activities were collected and preprocessed to handle complexities such as product assembly. The authors applied heuristic and inductive miner algorithms to discover the actual process models, revealing deviations and inefficiencies, particularly in certain stations and door usage. Discrete-event simulation was then used to evaluate different scenarios, including resource allocation and layout changes, to optimize warehouse performance. The study highlights the benefits of combining process mining with simulation for understanding and improving automated material handling processes, offering a data-driven approach to warehouse process analysis and optimization.

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Relevant	Reason	Summary
TRUE	The paper directly addresses the application of process discovery techniques (heuristic miner and inductive miner algorithms) in an automated material handling environment, specifically a cross-docking warehouse system. It details the structuring and preprocessing of event logs collected via RFID systems, providing insights into data gathering and event log transformation for effective process mining (L1, L2). The study discusses challenges such as deviations in process execution, data quality issues, and the complexity of integrating multiple product flows, which relate to common challenges in large-scale intralogistics systems (L3). Furthermore, the paper demonstrates how process discovery combined with discrete-event simulation can provide actionable insights for improving warehouse layout, resource allocation, and operational transparency, thus enhancing decision-making and system understanding (L4). The comprehensive case study, methodology, and results make it highly relevant for the SLR focused on process mining in automated intralogistics.	This paper presents a case study on applying process mining and simulation to a newly established cross-docking warehouse system of a chain store. Using RFID technology, detailed event logs were collected and preprocessed to capture the flow of multiple product types through various warehouse stations. The authors applied heuristic and inductive miner algorithms to discover the actual process models, revealing deviations and inefficiencies such as process deviations at specific stations and underutilized doors. They then used discrete-event simulation to evaluate different scenarios, including resource allocation and door usage optimization. The study resulted in recommendations for warehouse layout improvements, resource reallocation, and process adjustments to enhance throughput and operational efficiency. The paper contributes a practical data-driven approach combining process mining and simulation to improve transparency and decision-making in automated material handling environments.

Appendix B

Paper Relevance Conformance Checking

Table B.1: Both relevance labels of the 9 random selected papers

Title paper	**Relevant researcher**	**Relevant GPT 4.1 mini**
Matching of Business Data in a Generic Business Process Warehousing [20]	TRUE	TRUE
Towards a unified vision of business process and organizational data [21]	TRUE	TRUE
Process Simulation as a Basis for Process Mining in Intralogistics [14]	TRUE	TRUE
Four Decades of Data Mining in Network and Systems Management [32]	FALSE	FALSE
Process mining in CSCW systems [33]	FALSE	FALSE
Detecting System-Level Behavior Leading To Dynamic Bottlenecks [18]	TRUE	TRUE
Architecture of process mining based business process optimization [34]	FALSE	TRUE
Predictive Performance Monitoring of Material Handling Systems Using the Performance Spectrum [17]	TRUE	TRUE
A process mining approach for clinical guidelines compliance: real-world application in rectal cancer [35]	FALSE	FALSE

Table B.2: Confusion matrix classification relevant in full-text screening

		Predicted		
		Relevant	Not relevant	Total
Actual	Relevant	5	0	5
	Not relevant	1	3	4
	Total	6	3	9

Appendix C

Paper Metadata Conformance Checking

Table C.1: Comparison of Publication Year Extraction (Researcher vs. GPT 4.1 mini)

Title paper	**Year (Researcher)**	**Year (GPT 4.1 mini)**
Exploration and Analysis of Undocumented Processes Using Heterogeneous and Unstructured Business Data [25]	2014	2014
Analysis of Warehouse Operations using Process Mining Techniques: A Case Study of Scientific Laboratory Equipment and Facilities [12]	2020	2020
Process Mining: A framework proposal for Pervasive Business Intelligence [36]	2013	2012
Detecting System-Level Behavior Leading To Dynamic Bottlenecks [18]	2020	2020
A Hybrid Approach to Process Mining: Finding Immediate Successors of a Process by Using From-To Chart [15]	2009	2009
PMCube: A data-warehouse-based approach for multidimensional process mining [23]	2016	2016

Table C.2: Accuracy of Publication Year Extraction

Metric	**Count / Value**
Total Papers	6
Correct Matches	5
Mismatches	1
Accuracy	**83.3%**

Table C.3: Papers Tested for Country Extraction

Title paper	**Accuracy of Country Extraction**
Exploration and Analysis of Undocumented Processes Using Heterogeneous and Unstructured Business Data [25]	100%
Analysis of Warehouse Operations using Process Mining Techniques: A Case Study of Scientific Laboratory Equipment and Facilities [12]	100%
Process Mining: A framework proposal for Pervasive Business Intelligence [36]	100%
Detecting System-Level Behavior Leading To Dynamic Bottlenecks [18]	100%
A Hybrid Approach to Process Mining: Finding Immediate Successors of a Process by Using From-To Chart [15]	100%
PMCube: A data-warehouse-based approach for multidimensional process mining [23]	100%

Appendix D

Paper Metadata Results

Table D.1: List of Papers by ID and Publication Date (retrieved by GPT 4.1 mini)

ID	Title	Publication Year
1	A Hybrid Approach to Process Mining: Finding Immediate Successors of a Process by Using From-To Chart [15]	2009
2	Analysis of Warehouse Operations using Process Mining Techniques: A Case Study of Scientific Laboratory Equipment and Facilities [12]	2020
3	Process Simulation as a Basis for Process Mining in Intralogistics [14]	2025
5	An Interactive Approach to Support Event Log Generation for Data Pipeline Discovery [27]	2022
6	Generic meta model of cumulative fact table for non-sequential processes [37]	2021
7	Matching of Business Data in a Generic Business Process Warehousing [20]	2018
8	Process Mining: A framework proposal for Pervasive Business Intelligence [36]	2012
9	Predictive Performance Monitoring of Material Handling Systems Using the Performance Spectrum [17]	2019
10	Architecture of process mining based business process optimization [34]	2006
11	Predictive Monitoring of Business Processes: A Case Study Using a Cumulative Fact Table Model [24]	2024
12	Exploration and Analysis of Undocumented Processes Using Heterogeneous and Unstructured Business Data [25]	2014

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ID	Title	Publication Year
14	Process Mining in Data Science: A Literature Review [19]	2019
21	Detecting System-Level Behavior Leading To Dynamic Bottlenecks [18]	2020
23	Business Intelligence Revisited [30]	2017
24	Towards a unified vision of business process and organizational data [21]	2020
27	Convergence of Business and Their Processes: A Data Analytics Approach [22]	2025
28	EVErPREP: Towards an Event Knowledge Graph Enhanced Workflow Model for Event Log Preparation [26]	2025
29	Simulation and process mining in a cross-docking system: a case study [9]	2024
31	Event-based modeling and simulation for optimizing order picking [38]	2022
33	Warehousing Process Mining Research Based on Petri Net [11]	2021
34	Analytical treatment of transport logistics business processes by the Process Mining technology [10]	2020
36	Repairing Event Logs with Missing Events to Support Performance Analysis of Systems with Shared Resources [28]	2020
40	Context Aware Process Mining in Logistics [16]	2017
41	PMCube: A data-warehouse-based approach for multidimensional process mining [23]	2016
43	A dynamic time warped clustering technique for discrete event simulation-based system analysis [39]	2015
44	Material movement analysis for warehouse business process improvement with process mining: A case study [8]	2015
45	Multidimensional process mining with PMCube explorer [40]	2015
46	Research and design of process data warehouse for business process assessment [29]	2013
47	Process mining techniques in conformance testing of inventory processes: An industrial application [13]	2013

Table D.2: Author, University, and Country by Publication ID (retrieved by GPT 4.1 mini), see Table D.1 for reference for each paper

ID	Author Name	Author University/Affiliation	Country
1	Eren Esgin	Middle East Technical University, Informatics Institute	TR
1	Pinar Senkul	Middle East Technical University, Computer Engineering Department	TR
2	Suwanan Montianrat	Siam University	TH
2	Parham Porouhan	Siam University	TH
2	Poohridate Arpasat	Siam University	TH
2	Prajin Palangsantikul	Siam University	TH
2	Kwanchai Kungcharoen	Siam University	TH
2	Wichian Premchaiswadi	Siam University	TH
3	Jürgen Wunderlich	University of Applied Sciences Landshut	DE
3	Markus Schmidtner	University of Applied Sciences Landshut	DE
3	Theresa Götz	University of Applied Sciences Landshut	DE
3	Jonathan Schuster	University of Applied Sciences Landshut	DE
3	Florian Bross	Bross Consulting GmbH	DE
5	Dario Benvenuti	Sapienza University of Rome	IT
5	Leonardo Falleroni	Sapienza University of Rome	IT
5	Andrea Marrella	Sapienza University of Rome	IT
5	Fernando Perales	JOT Internet Media	ES
6	Bôa Djavph YOGANGUINA	Polytechnic School (ESP)	SN
6	Camara Mamadou Samba	Polytechnic School (ESP)	SN
7	Gamal Kassem	German University in Cairo	EG
7	Klaus Turowski	Otto-von-Guericke-University - OVGU	DE
8	Teresa Guarda	Isla - Superior Institute of Languages and Administration	PT
8	Manuel Filipe Santos	Minho University	PT
8	Maria Fernanda Augusto	Isla - Superior Institute of Languages and Administration	PT
8	Carlos Silva	Isla - Superior Institute of Languages and Administration	PT
8	Filipe Pinto	Polytechnic Institute of Leiria	PT
9	Vadim Denisov	Eindhoven University of Technology	NL
9	Dirk Fahland	Eindhoven University of Technology	NL

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ID	Author Name	Author University/Affiliation	Country
9	Wil M.P. van der Aalst	RWTH Aachen	DE
10	TANG Hongtao	Zhejiang University of Technology	CN
10	CHEN Yong	Zhejiang University of Technology	CN
10	Lu Jiansa	Zhejiang University of Technology	CN
11	Bôa Djavph Yoganguina	Polytechnic School (ESP)	SN
11	Camara Mamadou Samba	Polytechnic School (ESP)	SN
12	Sebastian Pospiech	Cologne Intelligence GmbH	DE
12	Sven Mielke	Hochschule Weserbergland	DE
12	Robert Mertens	Hochschule Weserbergland	DE
12	Martin Pelke	Hochschule Weserbergland	DE
12	Kavitha Jagannath	Hochschule Weserbergland	DE
12	Michael Städtler	Hochschule Weserbergland	DE
14	Muhammad Faizan	Universiti Kuala Lumpur	MY
14	Razi Ahmed	Universiti Kuala Lumpur	MY
14	Anwer Irshad Burney	KASBIT University	PK
21	Zahra Toosinezhad	Eindhoven University of Technology	NL
21	Dirk Fahland	Eindhoven University of Technology	NL
21	Özge Köroğlu	Eindhoven University of Technology	NL
21	Wil M.P. van der Aalst	RWTH Aachen, Department of Computer Science	DE
23	Jianwen Su	UC Santa Barbara	US
23	Yan Tang	UC Santa Barbara	US
24	Andrea Delgado	Universidad de la Republica	UY
24	Daniel Calegari	Universidad de la Republica	UY
27	Cristiana Vieira	CCG/ZGDV ICT Innovation Institute, Campus de Azurém	PT
27	Vânia Sousa	CCG/ZGDV ICT Innovation Institute, Campus de Azurém	PT
27	António Vieira	ALGORITMI Research Centre, University of Minho, Campus de Azurém	PT
27	Maribel Yasmina Santos	ALGORITMI Research Centre, University of Minho, Campus de Azurém	PT
28	Peter Filipp	Fraunhofer Institute for Integrated Circuits	DE
28	Rene Dorsch	Fraunhofer Institute for Integrated Circuits	DE
28	Andreas Harth	Friedrich-Alexander University Erlangen-Nuremberg	DE
29	Sadaf Shams-Shemirani	Sharif University of Technology	IR

Continued on next page

ID	Author Name	Author University/Affiliation	Country
29	Reza Tavakkoli-Moghaddam	University of Tehran	IR
29	Alireza Amjadian	Dalhousie University	CA
29	Bahar Motamedi-Vafa	Kharazmi University	IR
31	Pasquale Legato	University of Calabria	IT
31	Massimiliano Matteucci	PAC 2000 A Società Cooperativa	IT
31	Rina Mary Mazza	University of Calabria	IT
33	Fei Yin	Wuhan University of Technology	CN
33	Jingjing Cao	Wuhan University of Technology	CN
33	Jie Chu	Wuhan University of Technology	CN
33	Shurchkova Olga	Wuhan University of Technology	CN
34	A A Popov	Reshetnev Siberian State University of Science and Technology	RU
34	S N Masaev	Siberian Federal University	RU
34	D A Edimichev	Siberian Federal University	RU
34	O V Pomolotova	Siberian Federal University	RU
36	Vadim Denisov	Eindhoven University of Technology	NL
36	Dirk Fahland	Eindhoven University of Technology	NL
36	Wil M. P. van der Aalst	Eindhoven University of Technology	NL
40	Till Becker	University of Bremen	DE
40	Wacharawan Intoyoad	University of Bremen	DE
41	Thomas Vogelgesang	University of Oldenburg	DE
41	Hans-Jürgen Appelrath	University of Oldenburg	DE
43	Michael Johnstone	Deakin University	AU
43	Vu Thanh Le	Deakin University	AU
43	James Zhang	Deakin University	AU
43	Bruce Gunn	Deakin University	AU
43	Saeid Nahavandi	Deakin University	AU
43	Doug Creighton	Deakin University	AU
44	Mahendrawathi ER	Institut Teknologi Sepuluh Nopember	ID
44	Hanim Maria Astuti	Institut Teknologi Sepuluh Nopember	ID
44	Ika Rakhma Kusuma Wardhani	Institut Teknologi Sepuluh Nopember	ID
45	Thomas Vogelgesang	University of Oldenburg	DE
45	H.-Jürgen Appelrath	University of Oldenburg	DE
46	Hui Xia	Shan Dong University	CN
46	Qing Yao	Shan Dong University	CN
46	Fei Gao	Equipment Department of 71146 Army	CN

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ID	Author Name	Author University/Affiliation	Country
47	Zbigniew Paszkiewicz	Poznań University of Economics	PL

Appendix E

Paper Relevance Results

Table E.1: Overview of all the papers: relevance labels, reason and summary (provided by ChatGPT 4.1 mini)

Title	Authors	Relevant	Reason	Summary
Simulation and process mining in a cross-docking system: a case study [9]	Shams-Shemirani, S.; Tavakkoli–Moghaddam, R.; Amjadian, A.; Motamedi-Vafa, B.	True	The paper directly addresses RQ L1 by applying process discovery techniques (heuristic miner and inductive miner algorithms) to a cross-docking warehouse environment, which is a form of automated material handling and intralogistics. It also answers RQ L2 by detailing the structuring and preprocessing of event logs collected via an RFID system, including data integration for assembled products and event log enhancement for process mining. The paper discusses challenges (RQ L3) such as deviations in process execution, data quality issues, and the complexity of merging product flows, which affect process mining and simulation accuracy. For RQ L4, the study provides insights into process deviations, resource utilization, bottlenecks, and layout optimization, demonstrating how process discovery combined with simulation can improve transparency, system understanding, and decision-making in automated logistics operations. These contributions are supported by concrete methodology sections (Phases 1-5), detailed case study results, and simulation-based scenario analysis.	This paper presents a case study on applying process mining and discrete-event simulation to a newly established cross-docking warehouse system. Using RFID technology, the authors collect detailed event logs capturing product flows and activities. They preprocess and enhance the data to handle complex product assembly and bundling processes. Process discovery is performed using heuristic and inductive miner algorithms to reveal the actual process models and deviations from the intended workflows. The study identifies bottlenecks, resource usage patterns, and process variations. Subsequently, discrete-event simulation evaluates different scenarios, including resource allocation and door usage optimization, to recommend improvements in warehouse layout and operations. The integrated approach demonstrates how process mining combined with simulation can provide actionable insights for improving transparency and operational efficiency in automated intralogistics environments.

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Title	Authors	Relevant	Reason	Summary
Context Aware Process Mining in Logistics [16]	Becker, T.; Intoyoad, W.	True	The paper directly addresses RQ1 by applying process discovery techniques (process mining) in logistics and manufacturing environments, focusing on heterogeneous and dynamic processes. It contributes to RQ2 by proposing methods to preprocess and structure event logs using clustering (k-medoids) and context awareness (process frequency and cycle time) to improve process discovery outcomes. The methodology and results sections (3.3 and 3.4) detail distance measures and clustering approaches for handling complex event logs, which is relevant for structuring and preprocessing data. The discussion of challenges in managing heterogeneous data sources and the variability of processes relates to RQ3. Finally, the insights on identifying main processes and variants through clustering support RQ4 by enhancing transparency and understanding of process variations in logistics. The paper's focus on context-aware process mining and clustering to manage complexity and improve process documentation aligns well with the SLR scope on automated material handling and intralogistics.	This paper investigates the application of context-aware process mining in logistics and manufacturing, where processes are highly dynamic and heterogeneous. It proposes enhancing process discovery by incorporating contextual information, such as process frequency and cycle time stability, into clustering algorithms (k-medoids) to group similar process variants and identify main process patterns. The study evaluates different distance measures for quantifying process similarity and demonstrates that selecting frequent processes as cluster centers improves clustering quality. Using six real-world datasets, the paper shows how context-aware clustering can reduce complexity in process mining results, aiding automated process documentation and understanding in logistics environments. The work highlights challenges in handling heterogeneous event logs and suggests future research directions for further contfahland$_{i}nferring_{2}022extintegrationandclusteringimprovements$

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Title	Authors	Relevant	Reason	Summary
Architecture of process mining based business process optimization [34]	H. Tang; Y. Chen; Lu Jiansa	True	The paper directly addresses process mining techniques and their application in business process optimization, which aligns with the SLR's focus on process discovery in automated material handling and intralogistics environments. It discusses how event logs and data from business process systems (e.g., ERP, workflow management systems) can be extracted, modeled, and used for process mining, which relates to RQ L2 about structuring and preprocessing event logs. The proposed architecture includes data modeling, knowledge representation, and process mining methods to extract hidden knowledge from noisy and incomplete data, relevant to challenges in large-scale systems (RQ L3). Furthermore, the paper highlights how process mining can provide insights for process optimization at both macro (structure) and micro (parameter) levels, supporting transparency and decision-making improvements (RQ L4). Although the paper is not specific to warehouses or intralogistics, its general architecture and methodology for process mining-based optimization are applicable and informative for such environments, thus contributing to RQ L1 by describing applicable process discovery techniques.	This paper presents an architecture for business process optimization based on process mining. It reviews existing business process optimization methods and introduces process mining as a technique to extract process models and hidden knowledge from event logs of business systems like ERP and workflow management. The architecture includes components for data extraction, process modeling, knowledge discovery, and performance measurement, enabling optimization at both structural and parameter levels using case-based reasoning, simulation, and linear programming. Key challenges such as data modeling, knowledge representation, and handling noisy data are discussed. The paper emphasizes process mining's role in continuous monitoring and optimization of business processes, providing a foundation for improved transparency and decision-making.

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Title	Authors	Relevant	Reason	Summary
Analysis of Warehouse Operations using Process Mining Techniques: A Case Study of Scientific Laboratory Equipment and Facilities [12]	S. Montianrat; P. Arpasat; K. Kungcharoen; P. Porouhan; P. Palangsantikul; W. Premchaiswadi	True	The paper directly addresses L1 by applying process discovery techniques (using the DISCO tool) to warehouse operations involving scientific laboratory equipment, demonstrating practical use of process mining in a warehouse context. It also contributes to L2 by detailing how event log data from warehouse systems was extracted, structured (CSV format), and preprocessed for process mining analysis, including attribute mapping (case, activity, timestamp, resource). The methodology and results sections provide insights into challenges related to process variability and deviations in QC inspection times, which relate to L3's focus on challenges in applying process mining in operational settings. Finally, the paper offers insights (L4) into how process discovery revealed inefficiencies and variability in QC inspection times, suggesting potential improvements in warehouse workflow and quality control processes, thus enhancing transparency and decision-making. These connections are evident in sections III (Research Methodology), IV (Results and Conclusion), and the use of DISCO for process model visualization and analysis.	This paper presents a case study applying process mining techniques to analyze warehouse operations in a company distributing scientific laboratory equipment. Using event log data collected over one month, the authors preprocess and import data into the DISCO process mining tool to discover and visualize actual warehouse process flows. The study identifies significant variability in quality control inspection times among technicians, some shorter than organizational standards, potentially impacting product quality. The findings highlight how process mining can uncover operational inefficiencies and support improvements in warehouse procedures. The paper contributes practical insights into data preparation, process discovery application, and analysis of warehouse workflows using process mining.

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Title	Authors	Relevant	Reason	Summary
Process Simulation as a Basis for Process Mining in Intralogistics [14]	J. Wunderlich; T. Götz; F. Bross; M. Schmidtner; J. Schuster	True	The paper directly addresses multiple research questions relevant to the SLR. For L1, it discusses the application of process mining techniques on simulated intralogistics processes, specifically using discrete-event simulation combined with process mining tools (Sections II, IV). For L2, it details how event logs are generated from simulation models, including structuring and preprocessing of event data for effective process mining (Section IV.A). For L3, it identifies challenges such as complexity of logistics processes, limitations of traditional process mining (e.g., directly-follows graph issues), and the need for object-centric process mining to handle multiple interacting objects (Sections IV.D, V). For L4, it demonstrates insights gained from process discovery such as identifying bottlenecks, deviations, inefficiencies (e.g., long waiting times, non-compliance with FIFO, duplicate material provision), and how these insights can guide optimization and improve transparency and decision-making (Sections IV.C-E, V). The paper's methodology, case study, and discussion provide concrete examples and reflections highly relevant to automated material handling and intralogistics environments.	This paper explores the synergy between process simulation and process mining in intralogistics, proposing simulation as a data source for process mining to overcome challenges in real-world data availability and privacy. Using discrete-event simulation with Plant Simulation software, the authors model a simplified intralogistics system comprising storage, delivery and assembly, and shipping processes. Event logs generated from the simulation are analyzed with process mining tools to discover process models, identify deviations, bottlenecks, and inefficiencies such as long storage times, non-compliance with FIFO, and delays in assembly and shipping. The study highlights challenges in process mining for complex logistics, including limitations of traditional discovery techniques and the need for object-centric approaches. The findings demonstrate how process mining on simulated data can provide detailed insights into intralogistics processes, supporting transparency, system understanding, and targeted optimization. The paper concludes that combining simulation and process mining is a promising approach for data-driven process optimization in automated material handling environments.

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Title	Authors	Relevant	Reason	Summary
Matching of Business Data in a Generic Business Process Warehousing [20]	G. Kassem; K. Turowski	True	The paper addresses the structuring and integration of business process data into a multidimensional data warehouse model, which directly relates to RQ L2 about how event logs, system logs, or sensor data can be structured and preprocessed for effective process discovery. It proposes a generic meta-model for business processes that captures workflow models, cases, instances, activities, and business objects, enabling dynamic adaptation to process changes and detailed analysis. The paper also discusses challenges related to granularity and data consistency, which connect to RQ L3 on challenges in applying process mining to large-scale systems. Furthermore, the implementation of an ETL process and a business process search engine for analyzing business data supports RQ L4 regarding insights for transparency and decision-making. Although the paper is not specific to automated material handling or intralogistics, its generic approach to business process warehousing and process data modeling is applicable and valuable for the SLR's focus on automated logistics environments, where similar data structuring and analysis challenges exist.	This paper presents a generic conceptual meta-model and multidimensional data warehouse approach for integrating and analyzing business process data. It focuses on matching business data, including workflow models, cases, activities, and business objects, into a flexible data warehouse schema that supports dynamic process changes and detailed process analysis. The approach includes an advanced ETL process to extract and transform event logs into the warehouse and a business process search engine for ad-hoc querying and analysis. The paper discusses challenges such as granularity estimation and data consistency, and demonstrates a prototype implementation using SAP BW. The work aims to enable comprehensive business process intelligence beyond traditional OLAP by incorporating detailed business object attributes and their values, facilitating deeper insights into process behavior and supporting decision-making.

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Title	Authors	Relevant	Reason	Summary
Research and design of process data warehouse for business process assessment [29]	Xia, H.; Yao, Q.; Gao, F.	True	The paper directly addresses RQ L2 by discussing how heterogeneous process logs from different information systems can be structured, normalized, and preprocessed into a unified process data warehouse schema suitable for process mining. It proposes a process assessment-oriented process warehouse model (AOPWM) with a fact-dimension schema designed to store detailed process instance data, supporting effective process discovery and assessment. The paper also touches on challenges related to data heterogeneity and granularity, relevant to RQ L3. Furthermore, it demonstrates how process mining insights, such as efficiency, cost, and customer satisfaction metrics, can be derived from the warehouse data to support process transparency and decision-making, addressing RQ L4. Although the case study is in healthcare, the concepts and methodologies for process log structuring, warehouse design, and assessment are applicable to automated material handling and intralogistics environments where event logs and sensor data require integration and analysis for process understanding.	This paper presents the research and design of a process data warehouse tailored for business process assessment. It focuses on integrating heterogeneous process logs from various information systems into a unified, structured format based on the XES standard. The authors propose an access-oriented process warehouse model (AOPWM) using a fact-dimension schema to store granular process execution data, facilitating process mining and assessment. The paper includes a process assessment model that quantifies process instance quality through metrics like efficiency, cost, and customer satisfaction, enabling classification and decision support. A healthcare case study illustrates the application of the model and warehouse design for process analysis. The work contributes a structured approach to preprocessing and storing event logs to support process discovery and optimization.
Analytical treatment of transport logistics business processes by the Process Mining technology [10]	Popov, A.A.; Masaev, S.N.; Edimichev, D.A.; Pomolotova, O.V.	True	The paper directly addresses the application of process discovery techniques (heuristic miner, fuzzy miner) to event logs from a transport logistics warehouse, which aligns with RQ L1. It details the structuring and preprocessing of event logs with key attributes (activity, timestamp, case ID, resource), addressing RQ L2. The paper discusses challenges such as stray signals, missing timestamps, and data noise in large-scale event logs, relevant to RQ L3. It also demonstrates insights gained from process discovery, including identification of process deviations, bottlenecks, and recommendations for operational improvements, supporting RQ L4. The use of ProM software and detailed analysis of event sequences in a real transport warehouse context makes it highly pertinent to the SLR scope.	This paper presents an analytical study applying process mining techniques to the business processes of goods movement within a transport logistics warehouse. Using event logs collected from the warehouse, the authors employ heuristic and fuzzy mining algorithms via ProM software to build and analyze process models. The study identifies common and variant event sequences, detects data quality issues such as stray signals and missing timestamps, and performs conformance checking to reveal deviations from the ideal process flow. The research highlights practical challenges in event log preprocessing and demonstrates how process mining can uncover inefficiencies and errors in warehouse operations. Recommendations for improving process transparency and operational reliability are provided, along with suggestions for future work involving probabilistic modeling and network planning methods.

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Title	Authors	Relevant	Reason	Summary
EVErPREP: Towards an Event Knowledge Graph Enhanced Workflow Model for Event Log Preparation [26]	Filipp, P.; Dorsch, R.; Harth, A.	True	The paper directly addresses RQ L2 by proposing and demonstrating a structured workflow (EVErPREP) for event log preparation using Event Knowledge Graphs (EKGs), which enhances the structuring, semantic enrichment, and preprocessing of event logs from complex automated systems. The detailed workflow phases (exploration, extraction, processing) and the use of semantic querying, knowledge enrichment, consolidation, and aggregation provide concrete methodologies for handling complex event data, improving interpretability and reducing complexity. The case study on Munich Airport's Baggage Handling System, a highly automated intralogistics environment, provides practical evidence of the approach's effectiveness, thus also contributing to RQ L3 by discussing challenges like complexity and explainability in large-scale systems. Furthermore, the insights on improved explainability, reduced complexity, and enhanced data accessibility relate to RQ L4, showing how process discovery can improve transparency and decision-making. Although the paper does not focus on specific process discovery algorithms (L1), it supports the foundational data preparation critical for effective process mining in automated material handling contexts.	This paper introduces EVErPREP, a novel workflow model leveraging Event Knowledge Graphs (EKGs) to enhance event log preparation for process mining projects. It addresses the challenges of complex, knowledge-intensive event data preparation by using semantic web technologies to improve exploration, extraction, and processing of event data. The workflow consists of three phases: exploration of the EKG to identify relevant entities, semantic querying and knowledge enrichment to create annotated event logs, and processing steps including knowledge consolidation and semantic aggregation to reduce complexity and improve interpretability. The approach is evaluated through a case study on the Munich Airport Baggage Handling System, demonstrating improved explainability, reduced complexity, and better data handling in a complex automated intralogistics environment. The paper highlights how semantic enrichment and structured preprocessing can facilitate more effective process mining analyses in large-scale automated material handling systems.

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Title	Authors	Relevant	Reason	Summary
Multidimensional process mining with PMCube explorer [40]	Vogelgesang, T.; Appelrath, H.-J.	True	The paper directly addresses RQ L1 by presenting multidimensional process mining (MPM) techniques applied to event logs, enabling the discovery of multiple process models for subsets of cases with homogeneous features. It also contributes to RQ L2 by detailing how event logs can be structured as multidimensional data cubes (multidimensional event logs) and queried via OLAP to extract customized sublogs for process discovery. The discussion of challenges in interpreting numerous complex models and the proposed consolidation and visualization techniques relate to RQ L3, highlighting difficulties in managing complexity and model comparison in large-scale systems. Finally, the paper's case study and tool features demonstrate how process discovery insights can improve transparency and understanding of process variations, addressing RQ L4. Although the case study is in healthcare, the methodological contributions and tool capabilities are applicable to automated material handling and intralogistics environments where complex, variant-rich processes and large event logs are common.	This paper introduces PMCube Explorer, a novel tool for multidimensional process mining (MPM) that structures event logs as multidimensional data cubes and uses OLAP queries to extract sublogs for separate process discovery. It addresses the challenge of complex, variant-rich processes by enabling analysts to generate and compare multiple process models for homogeneous case groups. The tool includes features for consolidating and visualizing differences between models to aid interpretation. A case study on perioperative healthcare processes demonstrates the tool's ability to provide flexible, dynamic views on process variations and improve understanding of process behavior. The approach and tool are extensible and support various process discovery algorithms, offering a promising framework for analyzing complex event data in domains such as automated intralogistics.

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Title	Authors	Relevant	Reason	Summary
Warehousing Process Mining Research Based on Petri Net [11]	Yin, F.; Cao, J.; Chu, J.; Olga, S.	True	The paper directly addresses RQ L1 by applying the -algorithm for process discovery in a warehousing environment, specifically modeling the warehousing process using Petri nets. It also contributes to RQ L2 by detailing the structuring and preprocessing of event logs from an enterprise warehousing system, including conversion to XES format and filtering incomplete cases. For RQ L3, the paper discusses challenges such as discrepancies between predefined and actual processes, token mismatches during replay, and process flexibility, highlighting common issues in applying process mining to real intralogistics data. Regarding RQ L4, the paper demonstrates how process discovery provides insights into process deviations, bottlenecks, and opportunities for process improvement and optimization through consistency checking and Petri net model refinement. These contributions are supported by concrete methodology (Sections 3 and 4), experimental results (Section 4.3), and analysis of process optimization (Section 4.4).	This paper presents a study on applying process mining to the warehousing process of an enterprise using Petri nets. It uses the -algorithm implemented in the Prom tool to discover process models from event logs collected from a real warehousing system. The authors preprocess event logs by converting them to standard XES format and filtering incomplete cases. They build a theoretical Petri net model of the warehousing process and replay real event logs on this model to check consistency and fitness. The analysis reveals discrepancies and process variations, such as parallelism, cycles, and deviations from the predefined process. Based on these findings, the paper proposes improvements to the Petri net model to better reflect actual operations. The study highlights the value of process discovery for understanding complex warehousing workflows, identifying bottlenecks, and supporting process optimization in automated material handling environments.

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Title	Authors	Relevant	Reason	Summary
Event-based modeling and simulation for optimizing order picking [38]	Legato, P.; Matteucci, M.; Mazza, R.M.	True	The paper directly addresses the application of event-based modeling and simulation (event graphs) to the order picking process in a warehouse environment, which is a key part of intralogistics and automated material handling. It contributes to RQ L1 by demonstrating the use of event graphs as a process discovery and modeling technique that captures the event-driven logic of warehouse operations (Sections 3 and 4). It also relates to RQ L2 by discussing how event data and system states are structured and represented as event graphs, enabling detailed simulation and analysis (Section 3). The paper touches on challenges (RQ L3) implicitly by highlighting the complexity of modeling manual order picking with replenishment and dynamic order arrivals, and the need for detailed event-driven models to capture operational variability. For RQ L4, the paper provides insights into how process discovery via event graphs and simulation can improve transparency and decision-making by comparing order collection policies and their impact on throughput and resource utilization (Section 5). Although the focus is on manual order picking rather than fully automated systems, the methodology and findings are relevant for understanding and improving automated material handling processes in intralogistics. The integration of event graphs with process mining and digital twins mentioned in the abstract and introduction further supports its relevance to the SLR scope.	This paper presents an event graph (EG)-based modeling and simulation approach to optimize the manual order picking process in a warehouse operated by PAC 2000 A, a wholesale distribution cooperative in Italy. The EG formalism captures the event-driven logic of picking and replenishment activities, enabling detailed discrete-event simulation. The study compares current hour-free order collection policies with proposed time-window based policies, showing that the latter improves operational makespan and picker utilization. The paper emphasizes the use of EGs as a common ground for integrating simulation, digital twins, and process mining to enhance supply chain connectivity and decision support. The work includes a conceptual EG model, simulation implementation in Arena, and numerical results supporting managerial decision-making. Future directions include extending the model towards digital twin applications for real-time decision-making.

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Title	Authors	Relevant	Reason	Summary
Generic meta model of cumulative fact table for non-sequential processes [37]	B. D. Yoganguina; C. M. Samba	True	The paper directly addresses the structuring and preprocessing of event log data for process discovery in non-sequential processes, which is highly relevant to research question L2. It proposes a generic meta-model for cumulative fact tables that can capture all data related to process instances with loops, overcoming limitations of traditional cumulative fact tables designed for sequential processes. This contribution is crucial for handling complex event logs from automated material handling or intralogistics systems where loops and rework are common. The paper also discusses event log modeling and loop considerations (Section III), and the proposed meta-model (Section VI) supports improved data representation for process mining. While it does not focus specifically on intralogistics, the methodological insights on event log structuring and cumulative fact table design are applicable to large-scale automated warehouse environments, thus indirectly supporting L1 and L3 by enabling better data preparation and handling of complex process behaviors. The discussion on loops as explanatory variables for performance indicators also relates to L4 by highlighting how process discovery can reveal insights into delays and failures.	This paper investigates the limitations of traditional cumulative fact tables in data warehouses for capturing event log data of non-sequential business processes characterized by loops. It proposes a generic meta-model for cumulative fact tables that accommodates the recording of all data related to process instances with loops, enabling comprehensive event log representation. The work includes a review of sequential versus non-sequential process models, event log meta-modeling, and the design and filling procedures of cumulative fact tables. The proposed model supports better data structuring for process mining applications, particularly for processes with rework and repeated tasks. The paper concludes by emphasizing the importance of this approach for uncovering hidden process information and lays groundwork for future work on data warehouse design and process performance evaluation.

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Title	Authors	Relevant	Reason	Summary
Repairing Event Logs with Missing Events to Support Performance Analysis of Systems with Shared Resources [28]	Denisov, V.; Fahland, D.; van der Aalst, W.M.P.	True	The paper directly addresses challenges in process mining for automated material handling systems (MHS) with shared resources, which is highly relevant to intralogistics environments like Jumbo's distribution center. It contributes to RQ L1 by applying process discovery techniques that reconstruct missing events in event logs using synchronous proclets and linear programming, specifically tailored for warehouse-like systems with shared resources and FIFO queues. For RQ L2, it provides a detailed methodology for structuring and preprocessing incomplete event logs by inferring missing events and their timestamps, using a model-based approach that integrates control-flow, resource, and queue perspectives. Regarding RQ L3, the paper discusses challenges such as incomplete logging, many-to-many interactions between cases and resources, and the complexity of reconstructing event orders and timestamps in large-scale systems, proposing a solution that handles these via token trajectories and LP constraints. For RQ L4, the paper demonstrates that process discovery can restore event logs to enable accurate performance analysis, load estimation, and improved transparency into system behavior, even under incomplete data and irregular system performance. The evaluation on synthetic and real-life baggage handling system logs shows practical applicability and insight generation for performance monitoring and root cause analysis. These aspects are concretely discussed in sections 3 (modeling), 4 (problem formalization), 5 (methodology), and 6 (evaluation).	This paper presents a novel approach to repairing incomplete event logs in systems with shared resources, focusing on material handling systems such as airport baggage handling. It models the system using synchronous proclets to capture the interplay between process cases, resources, and FIFO queues, enabling decomposition into token trajectories. The approach reconstructs missing events and infers timestamp intervals through linear programming, ensuring consistency across process, resource, and queue perspectives. The method addresses challenges of incomplete logging and many-to-many case-resource interactions, which are common in automated intralogistics environments. Evaluation on synthetic and real-life data demonstrates that the approach can accurately restore event timestamps and support reliable performance analysis, including load estimation and detection of bottlenecks. This work advances process discovery techniques for automated material handling by improving event log completeness and enabling deeper insights into system dynamics and performance.

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Title	Authors	Relevant	Reason	Summary
Towards a unified vision of business process and organizational data [21]	A. Delgado; D. Calegari	True	The paper directly addresses RQ L2 by proposing a metamodel-based approach to integrate and unify process data (event logs) and organizational data (databases) from Business Process Management Systems (BPMS). It discusses structuring and preprocessing heterogeneous data sources, including event logs and organizational databases, to enable effective process discovery and data mining. The paper also touches on challenges relevant to RQ L3, such as data matching, data quality, and linking process events with organizational data, including timestamp synchronization issues in distributed systems. Furthermore, it contributes to RQ L4 by illustrating how the integrated data model can provide richer insights into process execution and organizational context, enabling advanced queries and combined analysis for improved transparency and understanding of business processes. The detailed example of a university student mobility process demonstrates practical application in an automated environment with complex data sources. Although the paper does not focus exclusively on automated material handling or intralogistics, its foundational approach to integrating process and organizational data is highly relevant and applicable to such environments, especially for improving visibility and understanding of complex automated systems like Jumbo's distribution center.	This paper proposes a unified metamodel and framework to integrate business process data (event logs from BPMS) and organizational data (databases and other information systems) to provide a comprehensive view of process execution. It addresses the challenge of linking heterogeneous data sources by defining a model-driven approach that captures both process and data perspectives, enabling the application of process mining and data mining techniques on a common data foundation. The authors discuss scenarios for intra- and inter-organizational processes, data matching techniques, and data quality considerations. An example case study involving a student mobility process at a university illustrates the practical implementation of the approach using Activiti BPMS and MySQL databases. The integrated model supports advanced querying and analysis, enhancing transparency and understanding of process behavior in complex environments. The work lays groundwork for future extensions to inter-organizational processes and integration with data warehouses and mining tools like ProM.

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Title	Authors	Relevant	Reason	Summary
Convergence of Business and Their Processes: A Data Analytics Approach [22]	Vieira, C.; Sousa, V.; Vieira, A.; Santos, M.Y.	True	The paper directly addresses RQ L1 by discussing the application of Object-Centric Event Logs (OCEL) in process mining, which is a recent and advanced process discovery technique relevant for complex environments like automated material handling. It also contributes to RQ L2 by proposing a meta-model (B&PADWM) for integrating business and process analytics within a Data Warehouse, detailing how event logs and process data can be structured and preprocessed for effective analysis. The paper touches on RQ L4 by demonstrating how process discovery insights can improve transparency and decision-making through a dashboard that links process inefficiencies to business performance, identifying bottlenecks and operational issues. Although it does not explicitly focus on intralogistics or automated warehouses, the demonstrated order management case and the general applicability of the meta-model to operational processes make it relevant. The paper does not explicitly discuss challenges in large-scale intralogistics systems (RQ L3), but its discussion on integration challenges between business and process analytics is indirectly related.	This paper proposes a novel meta-model (B&PADWM) that integrates business analytics and process analytics within a Data Warehouse system, leveraging Object-Centric Event Logs (OCEL) from process mining to provide a comprehensive understanding of business processes and their operational efficiency. It extends Kimball's dimensional design process to incorporate process discovery, enabling organizations to analyze both business performance and process effectiveness simultaneously. The approach is demonstrated through a case study on an order management process, where the meta-model supports the identification of bottlenecks and inefficiencies, linking process metrics to business KPIs via a dashboard. The work highlights the synergy between data science and process science to enhance decision-making and operational transparency, with potential applicability across various domains including automated material handling.

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Title	Authors	Relevant	Reason	Summary
Process Mining in Data Science: A Literature Review [19]	R. Ahmed; M. Faizan; A. I. Burney	True	The paper provides a comprehensive literature review of process mining techniques, including process discovery, conformance checking, and enhancement, which directly addresses RQ L1 about applied process discovery techniques. It discusses event logs, their structure, and preprocessing challenges (section 2.3), relevant to RQ L2. The paper also highlights challenges such as noise, complexity, and model fitness (sections 2.5 and 2.6), which relate to RQ L3. Furthermore, it elaborates on insights gained from process mining like bottleneck detection, conformance deviations, and operational support (sections 2.1, 2.6, 2.9), addressing RQ L4. Although the paper is general and not specific to intralogistics or automated material handling, it covers foundational concepts, techniques, and challenges essential for understanding and applying process mining in such environments. It also mentions applications in logistics and supply control (section 2.11.3), linking to the domain of interest. Therefore, it is useful for the SLR as a foundational and methodological reference.	This paper presents a broad literature review of process mining within data science, focusing on the extraction of process models from event logs through discovery, conformance checking, and enhancement techniques. It details various process discovery algorithms (e.g., Alpha Miner, Heuristic Miner), discusses event log characteristics and preprocessing challenges, and explains conformance checking methods including token-based replay and alignment. The paper also covers operational support aspects like detecting deviations, predicting outcomes, and recommending actions based on process mining results. Additionally, it reviews available process mining tools and categorizes process types (Lasagna and Spaghetti) with applications in business domains including logistics and supply chain. The work emphasizes the importance of fitness, precision, generalization, and simplicity as quality criteria for discovered models and highlights challenges in handling noise and complexity in event data.

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Title	Authors	Relevant	Reason	Summary
An Interactive Approach to Support Event Log Generation for Data Pipeline Discovery [27]	D. Benvenuti; L. Falleroni; A. Marrella; F. Perales	True	The paper directly addresses RQ L2 by proposing a structured, interactive approach to generate event logs through questionnaires and simulation when real event logs are unavailable or incomplete, which is common in complex automated systems. It also contributes to RQ L1 by enabling process discovery on simulated event logs representing data pipelines, a form of automated material handling workflows. The methodology (Sections III and IV) details how domain knowledge is captured and transformed into simulation models (e.g., BPMN, Declare) to produce event logs suitable for process mining. The preliminary evaluation (Section V) demonstrates practical applicability and validation with domain experts, highlighting challenges in data availability and human resource constraints, thus informing RQ L3. The discussion (Section VI) and conclusions emphasize how simulated event logs can provide insights into pipeline structure and behavior, supporting RQ L4 on improving transparency and understanding in automated processes. Although the focus is on data pipelines in digital marketing, the approach is general-purpose and applicable to intralogistics and automated material handling environments where event log extraction is difficult.	This paper presents an interactive, general-purpose approach to support event log generation for process discovery of data pipelines, especially when real event logs are unavailable or incomplete. It uses structured questionnaires to capture domain knowledge from business experts, which is then used to build simulation models (e.g., BPMN) that generate synthetic event logs. These logs feed process mining techniques to discover pipeline structures. The approach was preliminarily evaluated in a digital marketing context, showing its effectiveness and efficiency in producing realistic pipeline models and KPIs with limited human effort. The paper discusses challenges in event log extraction and highlights the benefits of simulated logs for understanding and analyzing complex workflows, proposing future work to validate and extend the method.

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Title	Authors	Relevant	Reason	Summary
Predictive Performance Monitoring of Material Handling Systems Using the Performance Spectrum [17]	V. Denisov; D. Fahland; W. M. P. van der Aalst	True	The paper directly addresses the application of process mining techniques, specifically process discovery and predictive process monitoring, in automated material handling systems (MHS), such as baggage handling systems in airports. It contributes to RQ L1 by introducing the Performance Spectrum (PS) as a novel process discovery and feature extraction technique that captures inter-case dependencies and performance dynamics over time in non-isolated cases typical of intralogistics environments. For RQ L2, the paper details how event logs from MHS can be structured into multi-channel performance spectra, aggregating event data into bins and classes to enable effective feature selection and preprocessing for machine learning-based process discovery and prediction. Regarding RQ L3, the paper discusses challenges such as non-stationarity, inter-case dependencies, and incomplete event logs in large-scale MHS, and proposes a methodology including domain knowledge and visual analytics for feature selection to overcome these. For RQ L4, the paper demonstrates that process discovery via the PS provides insights into workload prediction, bottleneck identification, and risk of delays, thereby improving transparency and decision-making in automated logistics operations. These connections are supported by detailed methodology (Sections III-V), real-life case studies and evaluations (Section VI), and discussions on challenges and insights (Sections VI-VII).	This paper presents a novel approach for predictive performance monitoring in automated material handling systems (MHS) using the Performance Spectrum (PS), a data structure that captures detailed and aggregated performance-related features from event logs over time and process steps. The authors formalize the Inter-Case Performance Prediction Problem (IC3P) to predict aggregate performance indicators in non-isolated, resource-sharing processes typical of MHS. They propose a methodology to structure event data into multi-channel PS, enabling feature extraction for machine learning models to predict workload and risk of delays in baggage handling systems. The approach is evaluated on both simulated and real event logs from a major European airport, showing improved prediction accuracy over state-of-the-art baselines. The paper also discusses challenges such as feature selection, non-stationarity, and incomplete logs, and highlights the potential of PS-based process discovery to enhance operational transparency and decision-making in intralogistics environments.

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Title	Authors	Relevant	Reason	Summary
Detecting System-Level Behavior Leading To Dynamic Bottlenecks [18]	Z. Toosinezhad; D. Fahland; Ö. Köroğlu; W. M. P. van der Aalst	True	The paper directly addresses RQ L1 by applying process discovery techniques to detect dynamic bottlenecks and system-level behaviors in a highly automated baggage handling system, a type of automated material handling environment. It proposes novel event abstraction levels (case-level, segment-level, system-level) and methods to preprocess and aggregate event logs and sensor data into meaningful system-level events, thus contributing to RQ L2. The study identifies challenges in detecting dynamic, temporal, and spatial bottlenecks that are not visible through traditional case-centric process mining, addressing RQ L3 by highlighting the complexity of correlating events across cases and segments in large-scale intralogistics systems. Finally, it demonstrates that process discovery can provide insights into bottleneck formation, propagation cascades, and frequent behavioral patterns that improve transparency and understanding of system dynamics, supporting RQ L4. These contributions are evidenced in sections III-VI (methodology on event abstraction, detection, and correlation), and VII (evaluation on real baggage handling data), with detailed discussion of challenges and insights in sections I, VIII.	This paper presents a novel approach to detect dynamic bottlenecks and the system-level behaviors that lead to them in highly automated material handling systems, specifically a baggage handling system at a major European airport. It introduces a multi-level event abstraction model (case-level, segment-level, system-level) to aggregate and analyze event logs, enabling the detection of temporal and spatial bottlenecks that traditional case-based process mining cannot identify. The approach includes methods for detecting high-load and blockage events, correlating these into cascades of system-level events, and mining frequent patterns within these cascades. The evaluation on seven days of real event data reveals thousands of dynamic bottlenecks and complex propagation patterns, providing causal explanations for performance issues. The study highlights challenges in event correlation and pattern support, and suggests future work on prediction and generalization to other domains.

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Title	Authors	Relevant	Reason	Summary
A Hybrid Approach to Process Mining: Finding Immediate Successors of a Process by Using From-To Chart [15]	E. Esgin; P. Senkul	True	The paper directly addresses L1 by proposing a novel process discovery technique that uses a from-to chart to find immediate successors in event logs, which is applicable to automated material handling and intralogistics environments. It also contributes to L2 by detailing how event logs can be structured into a FROMTOCHART table and how evaluation metrics (support, confidence, modified lift) are used to preprocess and filter transitions for effective process discovery. The paper discusses challenges related to noise and complexity in large datasets and proposes metrics to control model robustness and complexity, thus partially addressing L3. Finally, it provides insights into how process discovery can reveal activity sequences and relations, improving transparency and understanding of process flows, which relates to L4. The methodology is demonstrated on a real business process case, showing practical applicability of the approach.	This paper presents a hybrid process mining approach that adapts the industrial engineering tool 'from-to chart' to discover immediate successor relations among activities from event logs. It introduces evaluation metrics inspired by association rule mining to filter and assess transitions, enabling the construction of an optimized activity sequence that minimizes backtracking and irregular flows. The approach is applied to a real-world travel management process, demonstrating its ability to accurately identify immediate successors and generate a dependency/frequency graph representing the process model. The study emphasizes controlling model complexity and robustness when handling large event logs and suggests future work to extend the method to detect branching and loops in processes.

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Title	Authors	Relevant	Reason	Summary
PMCube: A data-warehouse-based approach for multidimensional process mining [23]	Vogelgesang, T.; Appelrath, H.-J.	True	The paper presents PMCube, a data-warehouse-based approach for multidimensional process mining (MPM) that models event logs as multidimensional data cubes, enabling flexible OLAP-style querying and process discovery on sublogs. This directly addresses RQ L1 by applying process discovery techniques (e.g., Inductive Miner, Fuzzy Miner) to sublogs extracted from automated event data. It also contributes to RQ L2 by detailing how event logs and case/event attributes can be structured in a multidimensional event log (MEL) within a relational data warehouse schema, supporting filtering, aggregation, and projection for effective process discovery. The paper discusses challenges (RQ L3) such as complexity and interpretability of numerous process models generated from many cube cells, and proposes consolidation (clustering) and advanced visualization techniques to mitigate these issues. Finally, it provides insights (RQ L4) on how multidimensional process mining can improve transparency and understanding by enabling multiple perspectives on process variants, supported by a healthcare case study demonstrating the identification of process differences and bottlenecks. Although the case study is in healthcare, the methodological contributions on data modeling, querying, mining, and analysis support are applicable to automated material handling and intralogistics environments where complex, multidimensional event data and process variants exist.	This paper introduces PMCube, a novel approach for multidimensional process mining based on modeling event logs as multidimensional data cubes within a data warehouse. It enables flexible OLAP-style queries to extract sublogs defined by case and event attributes, which are then independently mined using established process discovery algorithms. To handle the complexity of many resulting process models, PMCube incorporates consolidation techniques such as clustering and advanced visualization methods including difference views. The approach is implemented in a prototype and evaluated in a case study analyzing perioperative healthcare processes, demonstrating its ability to reveal process variants and support exploratory analysis. The paper contributes a detailed data model, query operators, and analysis support mechanisms that facilitate understanding of complex, variable processes from large-scale event data.

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Title	Authors	Relevant	Reason	Summary
Predictive Monitoring of Business Processes: A Case Study Using a Cumulative Fact Table Model [24]	B. D. Yoganguina; C. M. Samba	True	The paper directly addresses L1 by discussing the application of predictive monitoring techniques, including process mining and machine learning (e.g., linear regression, recurrent neural networks) in a logistics context. It contributes to L2 by proposing and detailing the use of a cumulative fact table model for structuring and preprocessing event data from IoT sensors and transportation management systems, enabling real-time and cumulative process state tracking. The paper also touches on L3 by discussing challenges related to data consistency, scalability, and complexity in cumulative fact table models when applied to dynamic and large-scale processes. Finally, it provides insights relevant to L4 by demonstrating how predictive monitoring improves transparency, decision-making, and operational efficiency in logistics through accurate delivery time forecasts and anomaly detection. These connections are supported by the methodology (Section III), case study and results (Section IV), and discussion (Section V).	This paper explores the use of a cumulative fact table model within data warehouses to enhance predictive business process monitoring (BPPM), focusing on a logistics case study. It integrates process mining concepts with machine learning algorithms to predict delivery times and detect anomalies in real-time using event data from IoT sensors and transportation management systems. The study demonstrates high prediction accuracy and reduced anomaly detection times, enabling better operational planning and decision-making. The paper discusses advantages and limitations of the cumulative fact table approach, including data storage efficiency and challenges in handling complex, dynamic processes. It concludes that this approach offers a promising infrastructure for proactive process management in logistics and other sectors.

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Title	Authors	Relevant	Reason	Summary
A dynamic time warped clustering technique for discrete event simulation-based system analysis [39]	Johnstone, M.; Le, V.T.; Zhang, J.; Gunn, B.; Nahavandi, S.; Creighton, D.	True	The paper directly addresses RQ L2 by detailing how discrete event simulation (DES) output logs, consisting of event sequences and timestamps, can be structured and preprocessed for process discovery using dynamic time warping (DTW) to measure similarity between entity flows. It also contributes to RQ L3 by discussing challenges such as selecting the number of clusters (K) in k-medoids clustering and managing large, complex event logs with many unique paths. Furthermore, it informs RQ L4 by demonstrating how clustering of entity trajectories reveals insights into system behavior, bottlenecks, and causal factors affecting performance, thus improving transparency and decision-making in automated logistics systems. The methodology is applied to a real-world automated baggage handling system, which is analogous to automated material handling and intralogistics environments, making the findings transferable. Although the paper does not explicitly focus on process discovery techniques in warehouses (RQ L1), its approach using DTW and clustering on event logs from a complex automated system is highly relevant and complementary to process mining in intralogistics.	This paper presents a novel method combining dynamic time warping (DTW) and k-medoids clustering to analyze discrete event simulation (DES) output logs for understanding system behavior. The approach processes event sequences and timestamps of entities flowing through a system to measure similarity and group similar trajectories, revealing causal factors influencing overall system performance. Applied to a complex airport baggage handling system, the method identifies clusters of entity flows with distinct processing paths and dwell times, highlighting bottlenecks and inefficiencies. The technique offers a global view of system dynamics beyond traditional local process metrics, providing actionable insights for system control and improvement. The method is unsupervised, simulation-software agnostic, and has potential applications in agent-based modeling and business event log analysis, making it relevant for automated material handling and intralogistics contexts.
Material movement analysis for warehouse business process improvement with process mining: A case study [8]	ER, E.R.; Astuti, H.M.; Wardhani, I.R.K.	True	The paper directly addresses RQ1 by applying the Heuristic Miner algorithm for process discovery in a warehouse environment, specifically using SAP WM event logs. It also contributes to RQ2 by detailing the extraction and preprocessing of event logs from SAP tables LTAK and LTAP, including case ID generation and attribute selection for effective process mining. The paper discusses challenges related to event log extraction from ERP systems and noise handling, which relates to RQ3. Finally, it provides insights into process deviations, bottlenecks, and root causes in warehouse material movement, supporting RQ4 on how process discovery can improve transparency and decision-making in automated logistics operations.	This paper presents a case study on applying process mining to warehouse management in a manufacturing company using SAP WM module data. It extracts event logs from SAP database tables, preprocesses them, and applies the Heuristic Miner algorithm in PROM to discover the actual material movement process. The study identifies deviations from standard procedures, bottlenecks such as prolonged storage times and inefficient transfers, and root causes like insufficient capacity of forklifts and racks. The paper contributes practical recommendations for warehouse process improvement and enriches literature on process mining applications in warehousing contexts.

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Title	Authors	Relevant	Reason	Summary
Process mining techniques in conformance testing of inventory processes: An industrial application [13]	Paszkiewicz, Z.	True	The paper directly addresses L1 by applying process discovery and conformance checking techniques to inventory processes in a warehouse environment supported by a Warehouse Management System (WMS). It uses event logs from the WMS, detailing how these logs are structured and preprocessed (L2), including attributes like activity name, timestamp, pallet ID, and storekeeper info. The paper discusses challenges related to large event logs (over 700,000 events) and tool limitations, such as ProM's inefficiency with big data, which relates to L3. It also provides insights into process deviations, FIFO policy adherence, quality assurance, and work distribution, demonstrating how process discovery can improve transparency and decision-making in automated logistics operations (L4). The detailed case study, methodology, and tool evaluation make it highly relevant for the SLR.	This paper presents an industrial case study applying process mining techniques, specifically process discovery and conformance checking, to inventory processes managed by a Warehouse Management System in a mattress production company. It analyzes over 550,000 events related to product management activities such as production, storage, shipment, and damage handling. The study compares the discovered de facto process model with the normative de jure model using tools like ProM and Fluxicon Disco, revealing process variants, deviations, and adherence to rules like FIFO and quality assurance. It discusses data structuring from WMS logs, challenges in handling large-scale event data, and the complementary use of academic and commercial tools. The findings provide actionable insights for improving process transparency, compliance, and operational understanding in automated warehouse environments.

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Title	Authors	Relevant	Reason	Summary
Exploration and Analysis of Undocumented Processes Using Heterogeneous and Unstructured Business Data [25]	S. Pospiech; S. Mielke; R. Mertens; M. Pelke; K. Jagannath; M. Stadler	True	The paper addresses the challenge of discovering undocumented processes by integrating heterogeneous and unstructured business data sources, which is directly relevant to RQ L2 regarding structuring and preprocessing event logs and sensor data for process discovery. It proposes a formalism and prototype for tracing processes across multiple data stores, including unstructured documents, using components and connectors to reconstruct process instances. This approach is applicable to automated material handling environments where event logs may be incomplete or scattered. The paper also discusses challenges in handling unstructured data and mapping keys across systems, which relates to RQ L3 on challenges in large-scale intralogistics systems. Furthermore, the visualization and generalization of process traces provide insights into process transparency and understanding, addressing RQ L4. Although the paper does not focus specifically on warehouse or intralogistics domains, its methodology for process discovery from heterogeneous data is applicable and valuable for such environments, including the context of Jumbo's automated warehouse. The paper does not focus on specific process discovery algorithms (RQ L1) but complements that by focusing on data preparation and integration aspects critical for effective discovery.	This paper presents a novel approach to discover and analyze undocumented business processes by leveraging heterogeneous and unstructured data sources such as databases, emails, and documents. It introduces a formalism using components and connectors to trace process instances across multiple systems, enabling the reconstruction and visualization of real process flows even when classical event logs are unavailable. The approach includes handling unstructured data through an ontology-based search engine and supports expanding queries to multiple process instances. A prototype implementation demonstrates the feasibility of integrating structured and unstructured data for process discovery. The work highlights challenges in data integration, key mapping, and query formulation, providing a flexible framework to improve process transparency and understanding in complex organizational environments.

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Title	Authors	Relevant	Reason	Summary
Process Mining: A framework proposal for Pervasive Business Intelligence [36]	T. Guarda; M. F. Santos; M. F. Augusto; C. Silva; F. Pinto	True	The paper proposes a framework integrating process mining (PM) into pervasive business intelligence (PBI) systems, which is relevant to understanding how process discovery techniques can be applied in complex environments (L1). It discusses the collection and preprocessing of data (ETL processes) from organizational sources to support PM, addressing data structuring and preprocessing (L2). The paper also highlights challenges in understanding complex, distributed, and evolving processes, which relates to common challenges in large-scale systems (L3). Finally, it emphasizes how PM can improve decision-making by providing insights into actual process execution and deviations, supporting transparency and system understanding (L4). Although the paper is not specific to intralogistics or automated material handling, its conceptual framework and discussion of PM integration with BI systems provide valuable insights applicable to such environments, especially for improving visibility and decision support.	This paper presents a conceptual framework for integrating process mining into pervasive business intelligence systems to enhance organizational decision-making. It reviews business intelligence maturity levels and introduces pervasive BI as an evolution that delivers timely, integrated information across all organizational levels. The framework outlines four key processes: defining objectives, data collection (including ETL), analysis (pattern detection and process model restructuring), and dissemination of insights. The paper discusses process mining perspectives—discovery, conformance, and extension—and emphasizes the challenges of analyzing complex, distributed, and evolving processes. The proposed framework aims to enable dynamic, data-driven process analysis to support strategic and operational decisions, thereby contributing to competitive advantage.

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Title	Authors	Relevant	Reason	Summary
Business Intelligence Revisited [30]	J. Su; Y. Tang	True	The paper directly addresses RQ L1 by discussing process mining techniques and their limitations in handling data alongside activities, emphasizing the need for integrating data into process discovery. It contributes to RQ L2 by proposing a workflow log framework that structures event and data logs into a unified, queryable format, facilitating flexible ad hoc queries and process analytics. The paper also touches on RQ L3 by identifying challenges such as lack of standards for workflow logs, the complexity of log reconstruction, and the need for efficient query languages and indexing for large logs. For RQ L4, it highlights how process discovery enriched with data can improve transparency and understanding of business processes, enabling better decision-making and process improvement. These connections are evident in sections II (ETL and OLAP), III (Workflow Logs), and IV (Problems and Challenges), where the authors propose a new log-based framework and discuss its advantages and challenges in business process analytics, which is applicable to automated material handling and intralogistics environments where complex event and sensor data are prevalent.	This paper revisits business intelligence (BI) by integrating big data and machine learning with process mining, focusing on the limitations of traditional ETL-OLAP approaches and activity-centric process mining. It proposes a novel framework centered on workflow logs that unify event, activity, and data records into a structured, queryable format to support flexible, ad hoc process analytics. The authors discuss the evolution of process modeling from activity-focused to data-aware approaches and introduce a query language for workflow logs enabling rich temporal and data-centric queries. The paper identifies key challenges such as the lack of workflow log standards, the need for efficient log query optimization, and the integration of data into process mining for enhanced process understanding and improvement. This framework offers a promising direction for analyzing complex, data-rich processes like those in automated intralogistics systems.
Assessing agile software development processes with process mining: A case study [41]	Marques, R.; Mira da Silva, M.M.; Ferreira, D.R.	False	The paper focuses on applying process mining to assess Scrum agile software development processes using event logs from Jira Software, a case-handling system. It addresses process discovery techniques in the context of software development, specifically agile methodologies, and discusses challenges in analyzing unstructured event logs from such environments. However, it does not relate to warehouse, intralogistics, or automated material handling environments, nor does it address structuring or preprocessing of event logs from such systems, challenges specific to large-scale intralogistics systems, or insights for improving automated logistics operations. The methodologies, data sources, and domain context are distinct from those in automated material handling and intralogistics, making it irrelevant to the SLR research questions L1-L4.	This paper presents a case study applying process mining techniques to evaluate the implementation of Scrum agile software development practices in two IT projects managed with Jira Software. It uses event logs extracted via the Jira REST API and applies various process mining algorithms (-miner, heuristics miner, inductive miner, fuzzy miner) to discover process models, analyze control-flow, performance, and organizational perspectives, and identify Scrum roles and bottlenecks. The study highlights the challenges of mining unstructured agile processes, the limitations of event logs in capturing all Scrum practices, and the importance of validating findings with domain experts. It concludes that process mining can reveal useful insights about Scrum implementation but is limited by the data available in case-handling systems.

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Title	Authors	Relevant	Reason	Summary
Cross-course Process Mining of Student Click-stream Data - Aggregation and Group Comparison [42]	T. Hildebrandt; L. Mehnen	False	The paper focuses on process mining applied to student click-stream data from Moodle LMS to analyze learning behaviors and performance differences across courses. It details data extraction, preprocessing, aggregation, and cross-course comparison of event logs and course sections. However, it does not address any of the SLR research questions related to automated material handling or intralogistics environments, process discovery techniques in warehouses, structuring event logs from such systems, challenges in large-scale intralogistics process mining, or insights for improving automated logistics operations. The domain, data sources, and application context are entirely educational and unrelated to material handling or intralogistics.	This paper presents methods for extracting, preprocessing, and aggregating student interaction event logs from Moodle LMS to enable process mining analyses across multiple courses. It introduces a data pipeline for standardizing course section labels to facilitate cross-course process discovery and compares learning paths of higher- and lower-performing student groups. The study finds that higher-performing students engage more frequently and dynamically with course materials, with clearer patterns emerging when data is aggregated at the course section level. The work contributes a toolchain and insights for educational process mining but is focused exclusively on learning analytics rather than industrial or logistics process mining.

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Title	Authors	Relevant	Reason	Summary
A process mining approach for clinical guidelines compliance: real-world application in rectal cancer [35]	Savino, M.; Chilorio, G.; Masciocchi, C.; Capocchiano, N.D.; Lenkowicz, J.; Gottardelli, B.; Gambacorta, M.A.; Valentini, V.; Damiani, A.	False	The paper focuses on applying process mining techniques, specifically conformance checking and process discovery, to assess adherence to clinical guidelines in rectal cancer treatment using hospital event logs. It employs the Pseudo-Workflow formalism and pMineR software to translate clinical guidelines into computer-interpretable formats and analyze real patient treatment pathways. However, the domain is healthcare and oncology, not automated material handling or intralogistics. The paper does not address process discovery techniques in warehouse or intralogistics environments (L1), structuring or preprocessing of event logs from material handling systems (L2), challenges in large-scale intralogistics process mining (L3), or insights for improving transparency and decision-making in automated logistics operations (L4). Its focus on clinical guideline compliance and healthcare event logs makes it irrelevant for the SLR on process mining in automated material handling and intralogistics.	This study applies process mining to evaluate compliance with clinical guidelines for rectal cancer treatment in a real-world hospital setting. It translates European Society of Medical Oncology guidelines into a computer-interpretable format using Pseudo-Workflow formalism and uses pMineR software to perform conformance checking and process discovery on event logs from 453 patients. The analysis identifies deviations from guidelines due to internal protocols and provides visualizations to aid clinical understanding. The work demonstrates the utility of process mining for healthcare guideline adherence assessment but is focused on oncology treatment pathways rather than logistics or material handling systems.

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Title	Authors	Relevant	Reason	Summary
Process mining in CSCW systems [33]	W. M. P. van der Aalst	False	The paper focuses on the application of process mining techniques within the broad domain of Computer Supported Cooperative Work (CSCW) systems, covering structured and unstructured workflows, groupware, ERP systems, and case handling systems. It discusses the ProM framework and illustrates process mining on various CSCW-related systems such as Staffware, InConcert, Outlook, SAP R/3, and FLOWer. However, it does not specifically address automated material handling, warehouse, or intralogistics environments. It lacks discussion on event log structuring or preprocessing tailored to such environments, does not identify challenges specific to large-scale intralogistics systems, nor does it provide insights directly applicable to improving transparency or decision-making in automated logistics operations. The examples and discussions are centered on office, business process, and cooperative work systems rather than automated material handling or intralogistics.	This paper explores the use of process mining techniques in the context of Computer Supported Cooperative Work (CSCW) systems, which include a spectrum from structured production workflows to unstructured groupware systems. It introduces the ProM framework that supports various mining algorithms and demonstrates its application on several CSCW systems such as Staffware, InConcert, Outlook, SAP R/3, and FLOWer. The paper classifies CSCW systems along data/process-centric and structured/unstructured dimensions and discusses how process mining can extract process models, social networks, and other insights from event logs generated by these systems. It aims to inspire broader application of process mining beyond traditional workflow management systems.
Event log generation in a health system: A case study [43]	Remy, S.; Pufahl, L.; Sachs, J.P.; Böttinger, E.; Weske, M.	False	The paper focuses on event log generation from complex healthcare data warehouses, specifically for treatment processes of low back pain. It addresses challenges in event extraction, correlation, and abstraction using clinical guidelines and medical ontologies. While it provides detailed methodology and lessons learned on structuring and preprocessing event data (related to RQ L2), it does not discuss process discovery techniques applied in automated material handling or intralogistics environments (L1), nor does it address challenges or insights specific to large-scale intralogistics systems (L3, L4). The domain and context (healthcare) differ significantly from automated warehouse or intralogistics systems, limiting direct applicability to the SLR focused on automated material handling.	This case study investigates the generation of event logs from a large, complex healthcare data warehouse to enable process mining of treatment processes for low back pain. The authors develop a pipeline involving event extraction, correlation, and abstraction, leveraging clinical guidelines, medical vocabularies, and ontologies to structure and normalize event data. They discuss challenges such as defining case notions (patient-, visit-, encounter-based), handling data granularity, and ensuring data privacy. The study includes technical implementation details and validation with domain experts, concluding with lessons learned and future research directions on improving event log generation in healthcare.

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Title	Authors	Relevant	Reason	Summary
Understanding What Software Engineers Are Working on The Work-Item Prediction Challenge [44]	R. Lämmel; A. Kerber; L. Praza	False	The paper focuses on predicting software engineers' work items (such as code diffs) by integrating and analyzing diverse logs and tool usage within software development workflows. It addresses challenges of loosely integrated tools, concurrent workflows, and incomplete logging in software engineering environments. However, it does not discuss process discovery techniques applied in warehouse, intralogistics, or automated material handling contexts (L1), nor does it address structuring or preprocessing event logs from such environments (L2). It also does not cover challenges specific to process mining in large-scale intralogistics systems (L3) or insights for improving transparency and decision-making in automated logistics operations (L4). The domain focus is software engineering workflows, not automated material handling or intralogistics, and the methods and data types differ substantially from those relevant to the SLR's scope.	This paper presents the 'work-item prediction challenge' in software engineering, focusing on predicting what specific work item (e.g., code diffs) a software engineer is working on at any time. It highlights the complexity due to many loosely integrated tools, concurrent and long-running workflows, and incomplete logging. The authors describe a system developed at Facebook that uses heuristics and machine learning to integrate diverse logs (version control, code review, task management, tool usage) and predict work items with time-windowed event sequences. The paper discusses data models, heuristics for correlating events to work items, word-correlation learning, and lessons learned about event log integration, prediction architecture management, and validation challenges. It situates the work within related research in program comprehension, mining software repositories, process mining, and machine learning, emphasizing the need to combine these techniques to address the challenge.

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Title	Authors	Relevant	Reason	Summary
Temporal data mining and process mining techniques to identify cardiovascular risk-associated clinical pathways in Type 2 diabetes patients [45]	A. Dagliati; L. Sacchi; C. Cerra; P. Leporati; P. De Cata; L. Chiovato; J. H. Holmes; R. Bellazzi	False	The paper focuses on applying temporal data mining and process mining techniques to clinical healthcare data for Type 2 Diabetes patients, aiming to identify clinical pathways and stratify cardiovascular risk. It deals with heterogeneous clinical and administrative data, event log construction, and process discovery in a healthcare context. However, it does not address any of the SLR research questions related to automated material handling, intralogistics, or warehouse environments. There is no discussion of process discovery techniques applied in automated logistics, no treatment of event logs or sensor data from such environments, no mention of challenges in large-scale intralogistics systems, nor insights into improving transparency or decision-making in automated material handling. The domain and application context are entirely different from the SLR focus.	This paper presents a methodology combining temporal data mining and process mining to analyze complex clinical datasets of Type 2 Diabetes patients. It describes preprocessing heterogeneous clinical and administrative data into event logs, applying temporal abstractions to represent patient states and trends, and using the Heuristics Miner algorithm within the ProM framework to discover clinical pathways. The study stratifies patients by cardiovascular risk and analyzes glycemic state transitions to identify meaningful care patterns. The work demonstrates how process mining can extract interpretable models from healthcare data to support personalized care planning.
Business Intelligence and Security-The Process Mining Approach [46]	V. P. Mishra; J. Dsouza; L. Elezabeth; N. Tiwari; M. J. Kaur	False	The paper provides a broad overview of process mining concepts, business intelligence, and security applications, focusing mainly on general business process intelligence and network security monitoring through audit trails. It discusses process mining techniques, data extraction, and challenges in business environments but does not specifically address automated material handling, warehouse, or intralogistics systems. There is no detailed discussion on structuring or preprocessing event logs from such environments, nor on challenges or insights specific to large-scale intralogistics systems. The security focus on intrusion detection and audit trails is not directly related to process discovery for material flow or throughput improvement in automated warehouses like Jumbo's system.	This paper surveys the application of process mining within business intelligence and network security contexts. It explains process mining steps—discovery, conformance, and enhancement—and their role in analyzing event logs to improve business processes and detect security breaches. The paper highlights business process intelligence (BPI) for monitoring, analyzing, and optimizing business workflows, and discusses technical, interpretative, and practical challenges in applying process mining. It also covers the use of audit trails for real-time intrusion detection and securing business networks. Overall, the paper emphasizes the integration of process mining with business intelligence to enhance operational transparency and security in general organizational settings, without focusing on automated material handling or intralogistics environments.

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Title	Authors	Relevant	Reason	Summary
Implementing heuristic miner for information system audit based on DSS01 COBIT5 (Case study: CV Narnia distribution) [47]	Ing Tawakkal; A. P. Kurniati; G. A. A. Wisudiawan	False	The paper focuses on applying heuristic process mining (Heuristic Miner) to support information system audit based on the COBIT 5 DSS01 process in a distribution company. It primarily addresses IT process assessment and audit using event logs, questionnaires, and interviews, with emphasis on validating IT operational procedures and monitoring. The study does not involve warehouse, intralogistics, or automated material handling environments, nor does it discuss structuring or preprocessing event logs from such systems. It also does not address challenges specific to large-scale intralogistics systems or provide insights for improving transparency or decision-making in automated logistics operations. Therefore, it does not contribute to any of the SLR research questions L1-L4 focused on process discovery and mining in automated material handling or intralogistics contexts.	This paper presents a case study applying heuristic process mining to support information system audit based on the COBIT 5 DSS01 Manage Operations process at CV Narnia Distribution. It integrates process mining with traditional audit methods to improve data collection and validation for IT process assessment. The study uses event logs from the company's information system to discover and conformance-check operational processes against standard operating procedures. It identifies bottlenecks and resource access issues, and combines mining results with interviews and questionnaires to rate process capability. The work demonstrates how process mining can enhance IT audit effectiveness but is focused on IT governance rather than logistics or material handling systems.

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Title	Authors	Relevant	Reason	Summary
A new essential functions installed DWH in hospital information system: Process mining techniques and natural language processing [48]	Honda, M.; Matsumoto, T.	False	The paper focuses on the application of process mining and natural language processing within a hospital information system context, specifically analyzing clinical and patient-related event logs. It does not address automated material handling, warehouse, or intralogistics environments, nor does it discuss event log structuring or challenges specific to large-scale intralogistics systems. Although it applies process mining techniques (ProM, XESame) to event logs, these are clinical HIS logs rather than intralogistics or automated material handling data. Therefore, it does not contribute to any of the SLR research questions related to process discovery in automated logistics.	This paper presents the integration of process mining techniques and natural language processing (NLP) into a hospital information system's data warehouse to analyze clinical event logs and unstructured discharge summaries. Using tools like ProM and XESame, the authors extract and visualize clinical process flows from HIS event logs such as user logins and patient movements. The NLP tool 'iKnow' is employed to analyze unstructured clinical notes, enabling extraction of semantically similar terms without relying on a thesaurus. The study aims to improve hospital activity efficiency, system security, and patient service by enhancing visibility into clinical processes and data.

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Title	Authors	Relevant	Reason	Summary
Proposed a Solution Based on Functional Dependency Graphs for Transactional Fact Table Error Correction [49]	B. D. Yoganguina; C. M. Samba	False	The paper focuses on correcting errors in transactional fact tables within data warehouses using functional dependency graphs. While it discusses process mining in a general sense and the structure of data warehouses, it does not address process discovery techniques applied specifically in warehouse, intralogistics, or automated material handling environments (L1). It also does not provide insights on structuring or preprocessing event logs or sensor data from such environments for process mining (L2), nor does it discuss challenges in applying process mining to large-scale intralogistics systems (L3). Finally, it does not explore the types of insights process discovery can provide for improving transparency or decision-making in automated logistics operations (L4). The paper is primarily concerned with data warehouse schema integrity and error correction rather than process mining application or methodology in intralogistics contexts.	This paper addresses frequent errors in transactional fact tables of data warehouses, specifically the mismatch between primary keys, cumulative fact table measures, and dimension granularity. It proposes a solution based on functional dependency graphs to correct these errors. The paper provides an overview of business processes, data warehouse architectures, and types of fact tables (transactional, periodic, cumulative). The main contribution is a method to ensure consistency in transactional fact tables by modeling dependencies between dimensions such as time and date. The focus is on data warehouse schema correctness rather than on process discovery or mining in automated material handling or intralogistics systems.
Four Decades of Data Mining in Network and Systems Management [32]	K. Kalegele; K. Sasai; H. Takahashi; G. Kitagata; T. Kinoshita	False	The paper provides a comprehensive survey of data mining applications in network and systems management (NSM) over four decades, focusing on activities such as discovery, monitoring, analysis, reporting, and domain knowledge acquisition. While it extensively covers data mining techniques, preprocessing, challenges, and insights in NSM, it does not specifically address process discovery or process mining techniques in warehouse, intralogistics, or automated material handling environments. The paper discusses event log and system data mining broadly in IT and network contexts but lacks direct relevance to automated material handling systems or intralogistics. It does not provide concrete methods or case studies related to structuring event logs or sensor data for process discovery in such environments, nor does it discuss challenges or insights specific to large-scale intralogistics systems or warehouse process transparency. Therefore, it does not contribute meaningfully to the SLR research questions L1-L4.	This paper surveys four decades of data mining applications in network and systems management, covering key activities like asset discovery, monitoring, analysis (including prediction, detection, optimization, and causality), reporting, and domain knowledge extraction. It reviews various data mining techniques and algorithms applied to system logs, network traffic, and other NS-data sources, highlighting challenges in data preprocessing, integration, and interpretation. The paper also discusses measures of interestingness, ongoing research frontiers such as big data analytics, and the importance of domain knowledge. While comprehensive in the NSM domain, it focuses on IT and network systems rather than automated material handling or intralogistics.

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Title	Authors	Relevant	Reason	Summary
Streamlining Workflow Automation with a Model-Based Assistant [50]	A. Tuyishime; F. Basciani; A. Di Salle; J. L. Cánovas Izquierdo; L. Iovino	False	The paper focuses on workflow automation tools (WATs) and proposes a model-based assistant to help define and maintain workflows by transforming them into BPMN models, analyzing data dependencies, and detecting evolution/refactoring issues. While it discusses process modeling, validation, and evolution in the context of software workflows, it does not address process discovery techniques applied specifically in warehouse, intralogistics, or automated material handling environments (L1). It also does not cover structuring or preprocessing of event logs or sensor data from such environments for process mining (L2), nor does it discuss challenges related to large-scale intralogistics systems (L3). The insights provided relate to improving workflow definition and error detection in software automation platforms rather than transparency or decision-making in automated logistics operations (L4). Therefore, it does not contribute meaningfully to any of the SLR research questions.	This paper investigates challenges in defining and maintaining workflows in low-code Workflow Automation Tools (WATs) and proposes a model-based assistant that transforms workflows into BPMN models for analysis. It introduces analyzers to detect problematic data dependencies and evolution-induced refactorings, providing recommendations via annotations to improve workflow correctness and sustainability. The approach is demonstrated using a running example implemented in the N8N platform, highlighting how the assistant can identify errors and suggest fixes. The work aligns with process model enhancement in process mining but focuses on software workflow automation rather than physical intralogistics or material handling systems.
Revealing Cause-Effect Relations in Comorbidities Analysis Using Process Mining and Tensor Network Decomposition [51]	J. T. Matamalas; A. Arenas; A. Martínez-Ballesté; A. Solanas; C. Alonso-Villaverde; S. Gómez	False	The paper focuses on applying process mining combined with tensor network decomposition to analyze cause-effect relations in biomedical comorbidities data. It addresses the preservation of temporal dimensions in tensor decompositions and the modeling of disease progression as processes. However, it does not discuss or apply process discovery techniques in warehouse, intralogistics, or automated material handling environments (L1). It does not cover structuring or preprocessing of event logs or sensor data from such environments (L2). The challenges discussed relate to biomedical data and tensor decomposition, not intralogistics systems (L3). The insights provided pertain to understanding disease interactions and comorbidities, not transparency or decision-making in automated logistics operations (L4). Therefore, it does not contribute meaningfully to the SLR research questions focused on automated material handling and intralogistics.	This paper proposes a novel methodology combining time-preserving tensor network decomposition with process mining to analyze cause-effect relations in comorbidities using large-scale biomedical event data. It introduces new tensor decomposition techniques that maintain temporal information critical for process mining and models disease progression as multilayer complex networks. The approach aims to uncover hidden transitions and causal relations in disease interactions from extensive clinical records. The work is positioned at the intersection of tensor algebra, process mining, and network medicine, with applications in healthcare data analysis rather than logistics or material handling.

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Title	Authors	Relevant	Reason	Summary
Process Cube for Software Defect Resolution [52]	M. Gupta; A. Sureka	False	The paper focuses on applying process cube and OLAP operations to software defect resolution processes using event logs from software repositories (Issue Tracking System, Peer Code Review, Version Control System). It presents a multidimensional process mining framework for comparing software defect resolution variants, specifically analyzing control flow and time perspectives for different bug types (Performance and Security). However, it does not address any of the SLR research questions related to automated material handling or intralogistics environments. The data sources, domain context, process discovery techniques, preprocessing methods, challenges, and insights are all specific to software engineering and defect resolution, not warehouse or logistics systems. Therefore, it does not contribute to understanding process discovery techniques, event log structuring, challenges, or insights in automated material handling or intralogistics.	This paper introduces a novel framework combining OLAP process cubes with process mining to analyze and compare software defect resolution processes across multiple dimensions such as issue type, priority, and status. Using data from Google Chromium's software repositories (ITS, PCR, VCS), the authors extract and integrate event logs to create multidimensional process cubes. They apply OLAP operations (slice, dice, roll-up, drill-down) to generate sublogs for process mining, enabling comparison of process variants for different bug types (Performance vs. Security). The study discovers runtime process models, identifies bottlenecks, analyzes activity frequencies, transitions, and anti-patterns, and proposes metrics like Bottleneck Ratio to highlight inefficiencies. The work demonstrates how multidimensional process mining can provide deeper insights into software defect resolution workflows.